



Image Processing (303105381)

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UNIT-2

Image Enhancement

Outline

- Background of image Enhancement
- Basic gray level (intensity) Transformations,
- Histogram Processing,
- Enhancement using Arithmetic/Logic Operations,
- Fundamentals of Spatial Filtering,
- Smoothing and Sharpening Spatial Filters,
- Combining Spatial Enhancement Methods.

Outcome of this unit:

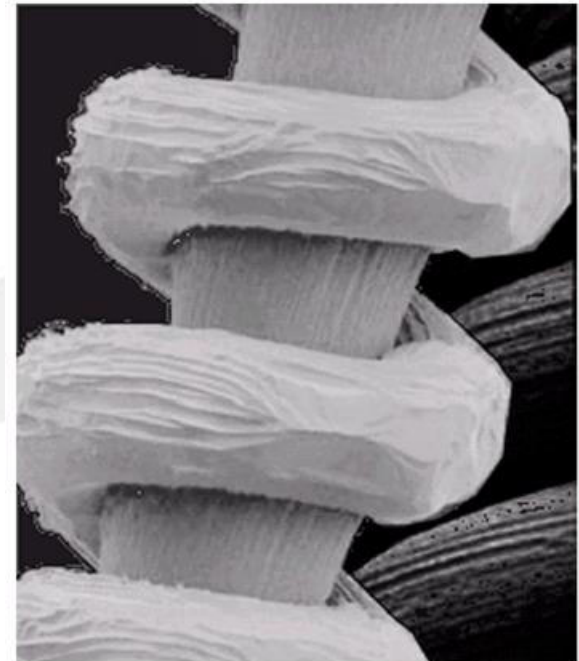
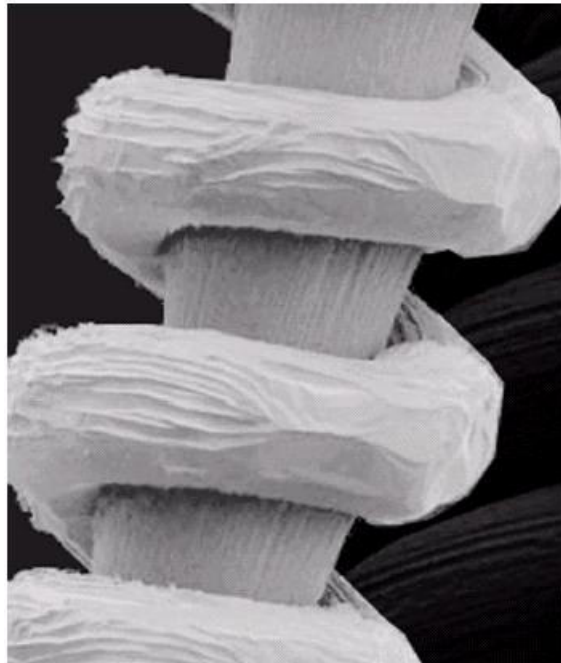
After completing the students will be able to:

- To understand the fundamental concepts of Image Enhancement.
- Apply the different image Enhancement techniques.
- Learn histogram and spatial filter techniques.

1. Background of Image Enhancement

Image Enhancement means improvement of images to be suitable for specific applications. Each image enhancement technique that is suitable for one application may not be suitable for other applications.

- Example:



Background

- A mathematical representation of **spatial domain enhancement**:

where $f(x, y)$: the input image

$g(x, y) := Tf(x, y)$:

$g(x, y)$: the processed image

T : an operator on f , defined over some neighborhood of (x, y)

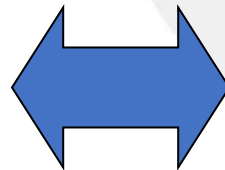
Spatial Domain

What is spatial domain :The space where all pixels form an image

In spatial domain we can represent an image by $f(x,y)$ where x and y are coordinates along x and y axis with respect to an origin

There is duality between Spatial and Frequency Domains

Images in the spatial domain are pictures in the xy plane where the word “distance” is meaningful.



Using the Fourier transform, the word “distance” is lost but the word “frequency” becomes alive.

continue

Goal: Image enhancement seeks

- To improve the visual appearance of an image, or
- Convert it to a form suited for analysis by a human or a machine.

Image enhancement does not, however,

- seek to restore the image, nor
- increase its information contents

Peculiarity:

- Actually, there is some evidence which suggests that a distorted image can be more pleasing than a perfect image!

Major Problem in Image Enhancement (IE):

- The lack of a general standard of image quality makes it very difficult to evaluate the performance of different IE schemes.
- Thus, Image Enhancement algorithms are mostly application-dependent, subjective and often adhoc.
- Therefore, mostly subjective criteria are used in evaluating image enhancement algorithms.

Subjective Criteria Used in Image Enhancement

A. Goodness scale: how good an image is

Overall goodness scale	Group goodness scale
Excellent (5)	Best (7)
Good (4)	Well above average (6)
Fair (3)	Slightly above average (5)
Poor (2)	Average (4)
Unsatisfactory (1)	Slightly below average (3)
	Well below average (2)
	Worst (1)

B. Impairment scale: how bad the degradation is in an image

- not noticeable (1)
- just noticeable (2)
- definitely noticeable but only slight impairment (3)
- impairment acceptable (4)
- somewhat objectionable (5)
- definitely objectionable (6)
- extremely objectionable (7)

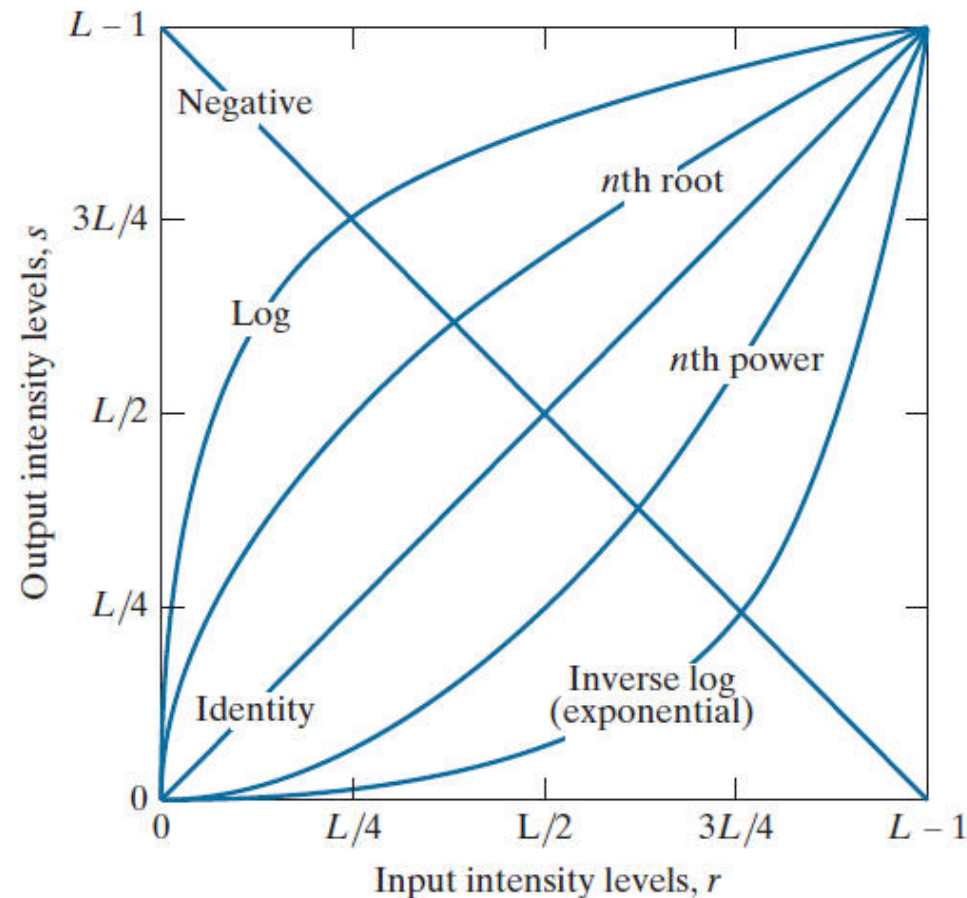
2. Basic Grey Level Transformations

Three of the most common are shown here

- Linear
 - Negative/Identity
- Logarithmic
 - Log/Inverse log
- Power law
 - n^{th} power/ n^{th} root

FIGURE

Some basic intensity transformation functions. Each curve was scaled *independently* so that all curves would fit in the same graph. Our interest here is on the *shapes* of the curves, not on their relative values.



Basic Spatial Domain Image Enhancement

Spatial domain technique operate directly on pixels of an image,
Most spatial domain enhancement operations can be reduced to the form

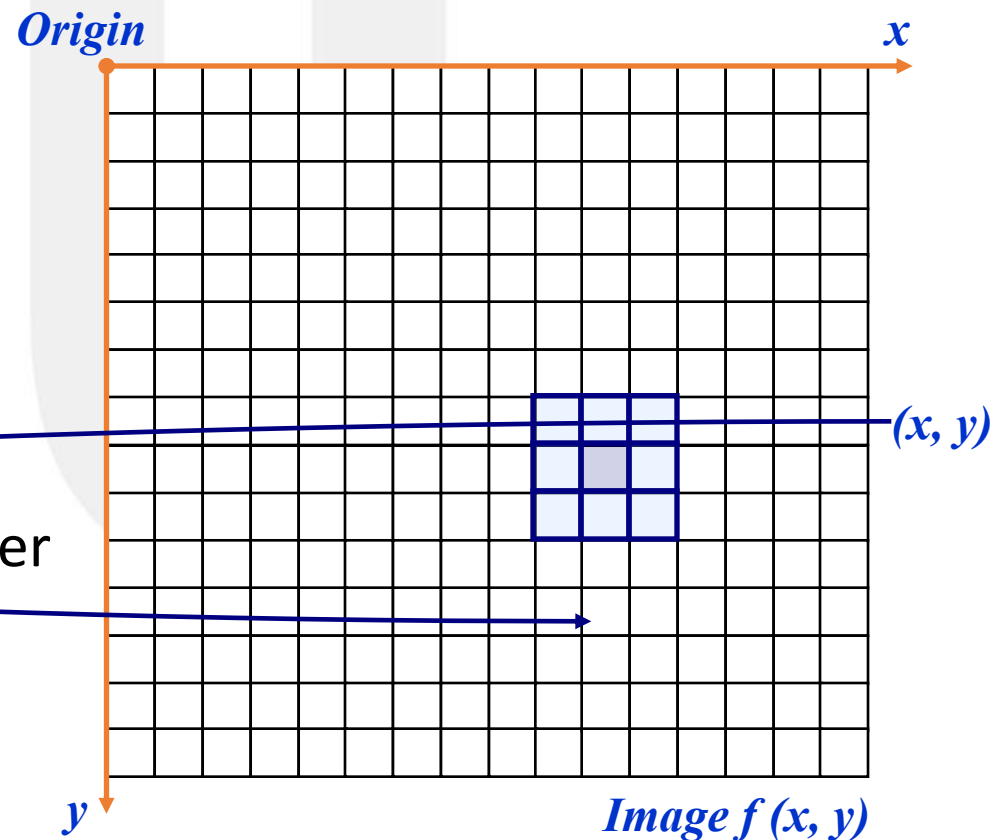
$$g(x, y) = T[f(x, y)]$$

where,

$f(x, y)$: input image,

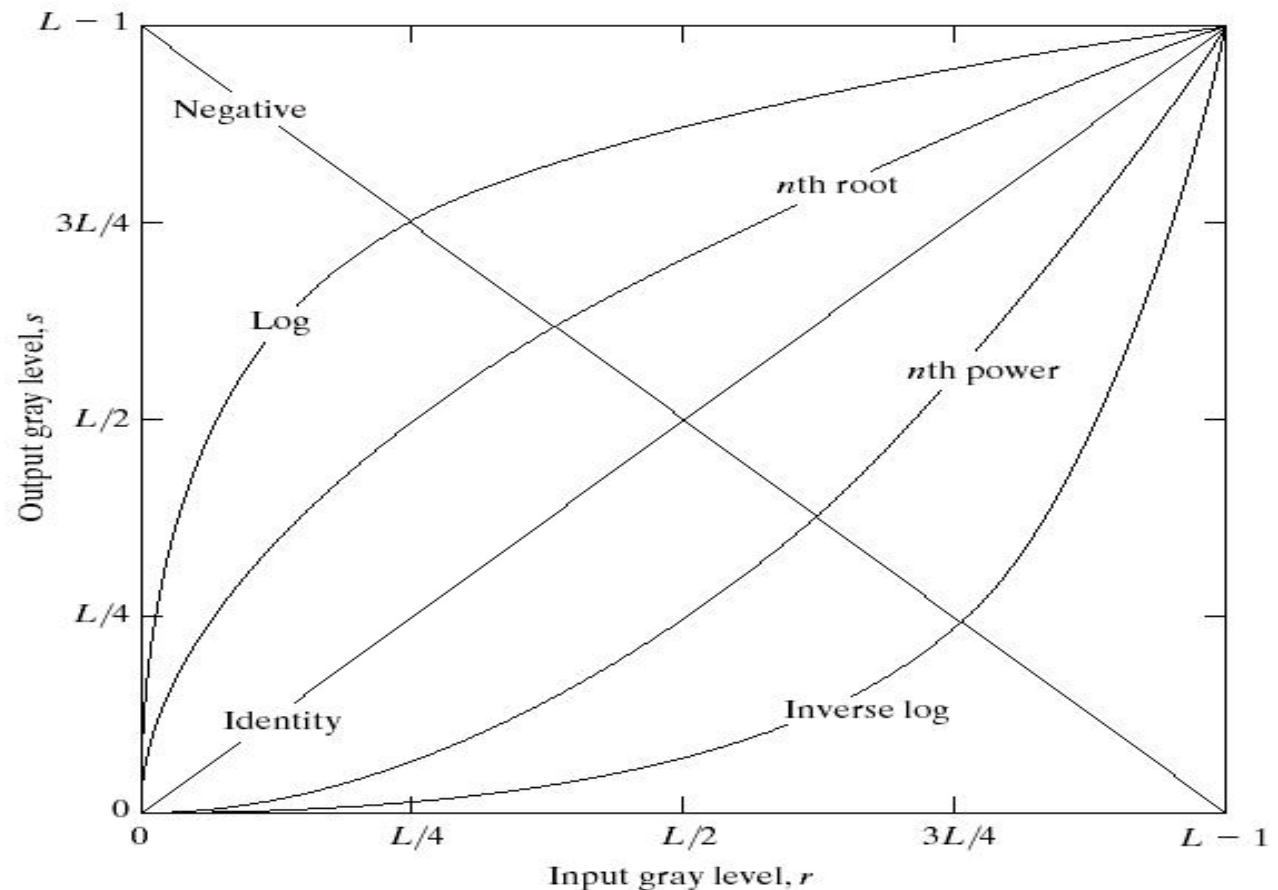
$g(x, y)$: the processed image
and

T : is some operator defined over
some neighbourhood of (x, y)



Some Basic Gray Level Transformations

FIGURE 3.3 Some basic gray-level transformation functions used for image enhancement.



Gray level transform

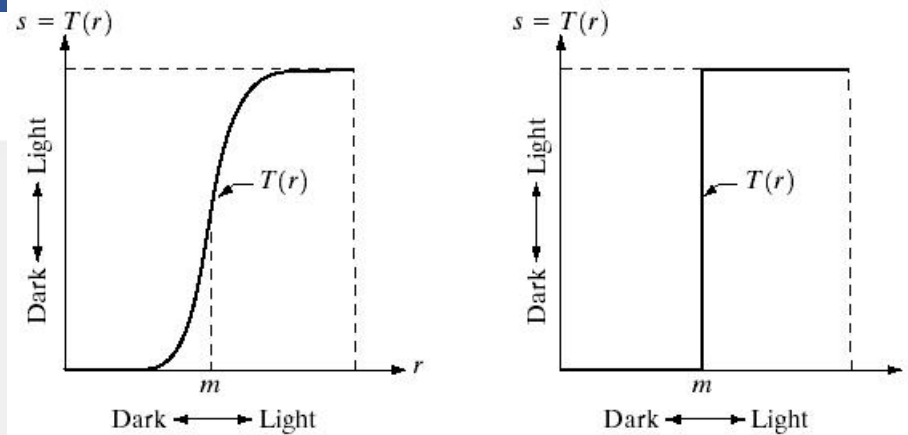
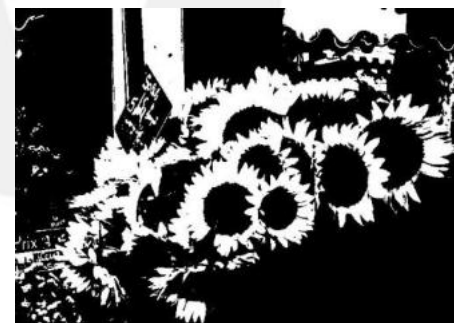


FIGURE 3.2 Gray-level transformation functions for contrast enhancement.

Point Processing

The simplest spatial domain operations occur when the neighbourhood is simply the pixel itself

In this case T is referred to as a *grey level transformation function* or a *point processing operation*

Point processing operations take the form

$$s = T(r)$$

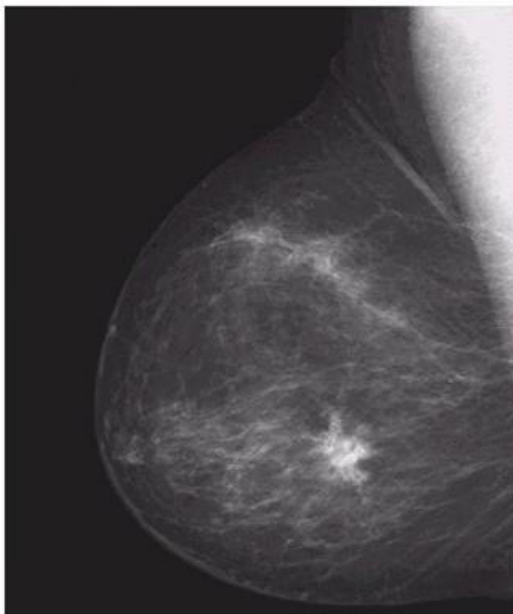
where s refers to the processed image pixel value and r refers to the original image pixel value

Point Processing Example: Negative Images

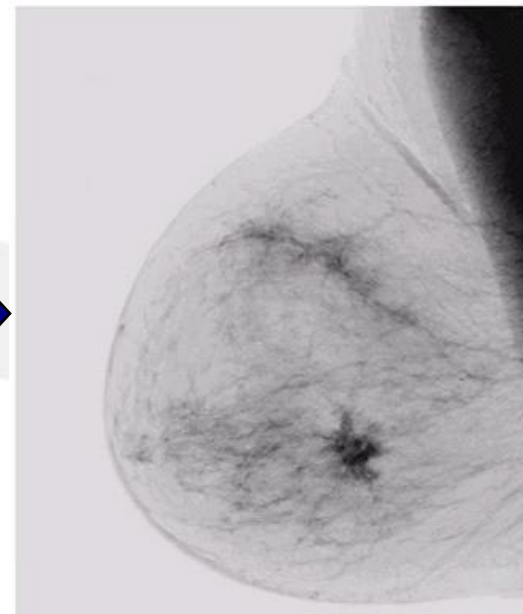
Negative images are useful for enhancing white or grey detail embedded in dark regions of an image.

Note: how much clearer the tissue is in the negative image of the mammogram below

Original
Image

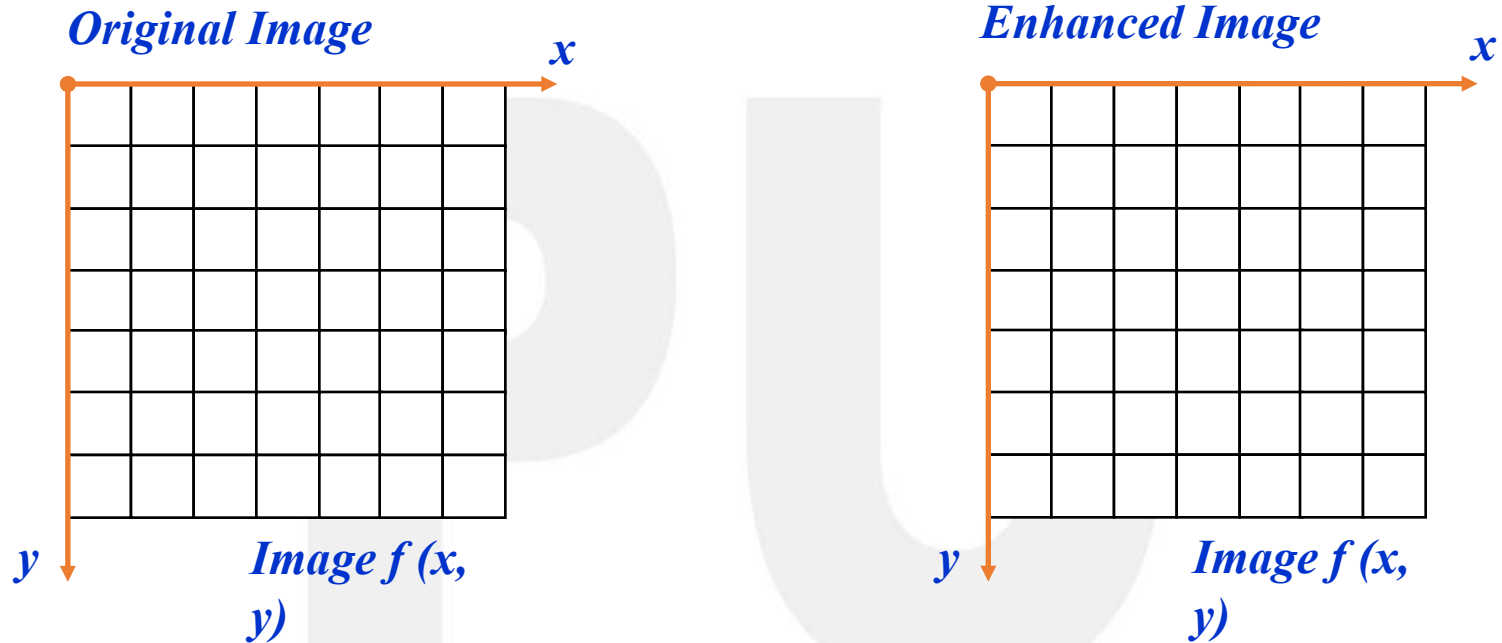


$$s = 1.0 - r$$



Negative
Image

Point Processing Example: Negative Images (cont...)



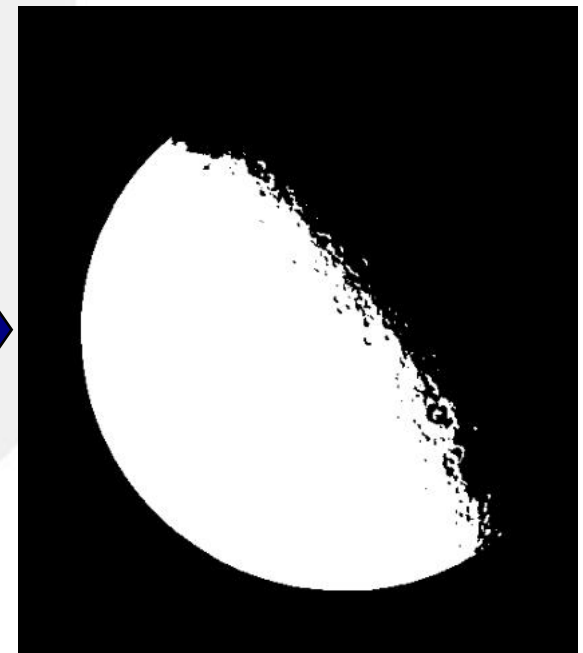
$$s = intensity_{max} - r$$

Point Processing Example: Thresholding

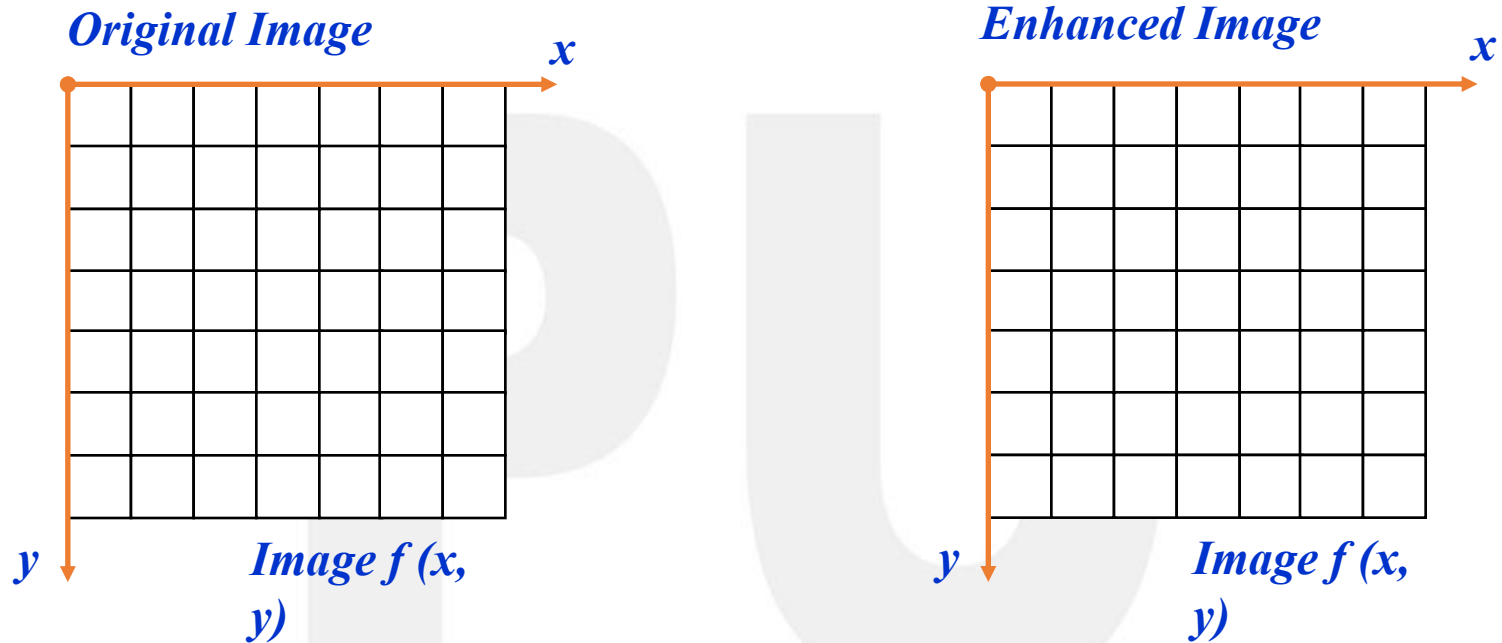
Thresholding transformations are particularly useful for segmentation in which we want to isolate an object of interest from a background



$$s = \begin{cases} 1.0 & r > \text{threshold} \\ 0.0 & r \leq \text{threshold} \end{cases}$$

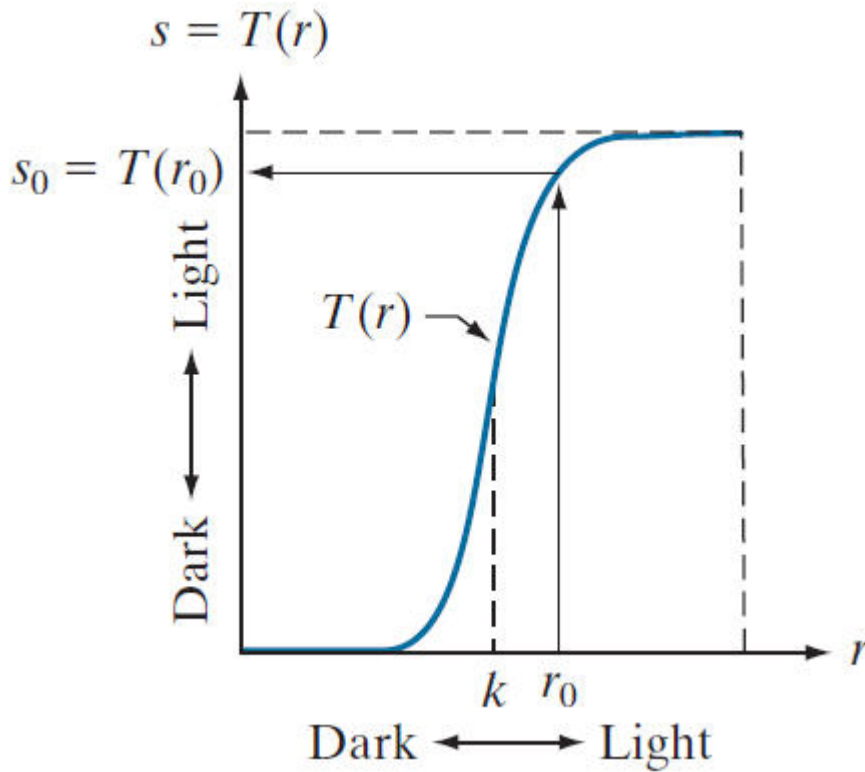


Point Processing Example: Thresholding (cont...)

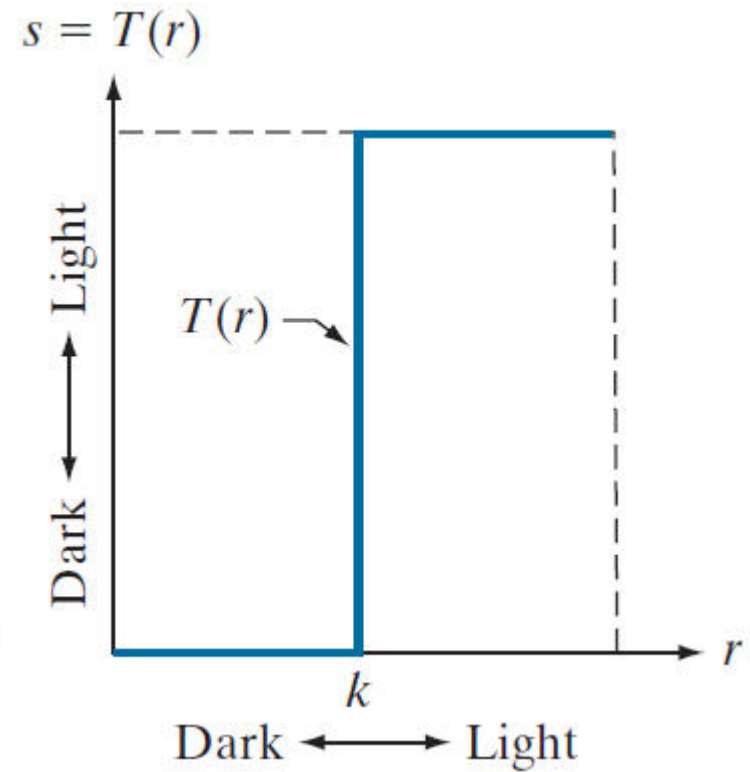


$$s = \begin{cases} 1.0 & r > \text{threshold} \\ 0.0 & r \leq \text{threshold} \end{cases}$$

Intensity Transformations



Contrast Stretching



Thresholding

Logarithmic Transformations

The general form of the log transformation is

$$s = c * \log(1 + r)$$

The log transformation maps a narrow range of low input grey level values into a wider range of output values

The inverse log transformation performs the opposite transformation.

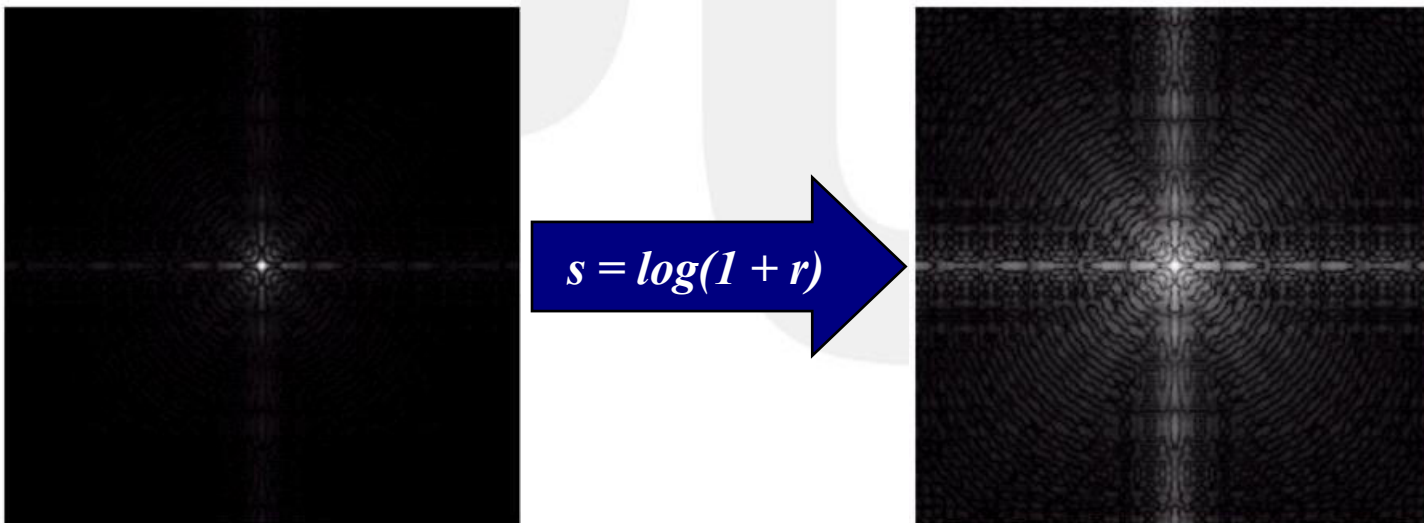
Log functions are particularly useful when the input grey level values may have an extremely large range of values

It maps a narrow range of low intensity values in the input into a wider range of output levels. The opposite is true for higher values of input levels.

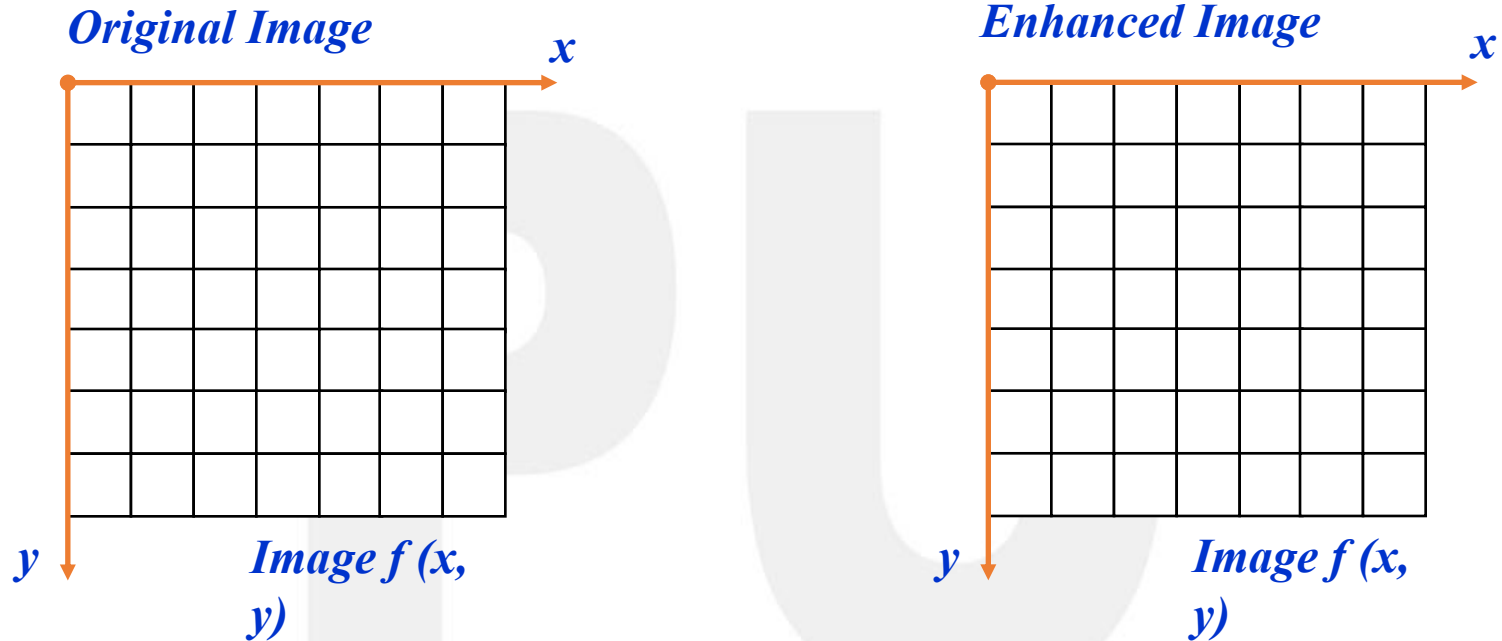
Logarithmic Transformations (cont...)

Generally used to expand the values of dark pixels in an image while compressing the higher level values.

In the following example the Fourier transform of an image is put through a log transform to reveal more detail



Logarithmic Transformations (cont...)



$$s = \log(1 + r)$$

We usually set c to 1

Grey levels must be in the range $[0.0, 1.0]$

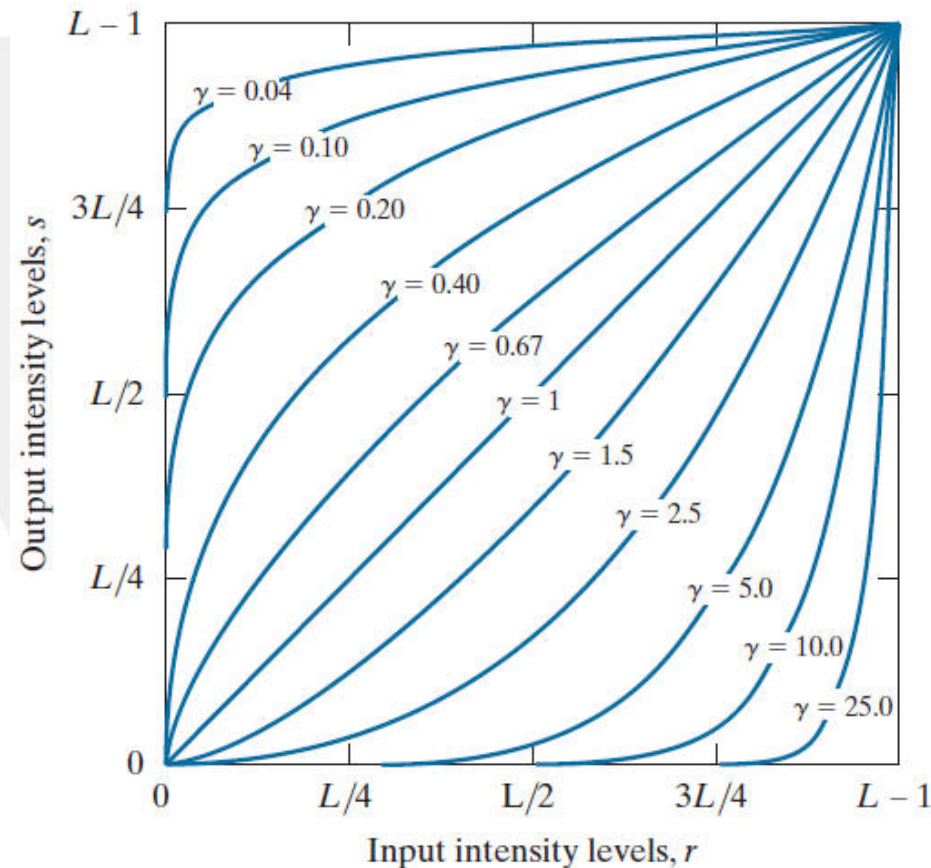
Power Law (Gamma) Transformations

Power law transformations have the following form

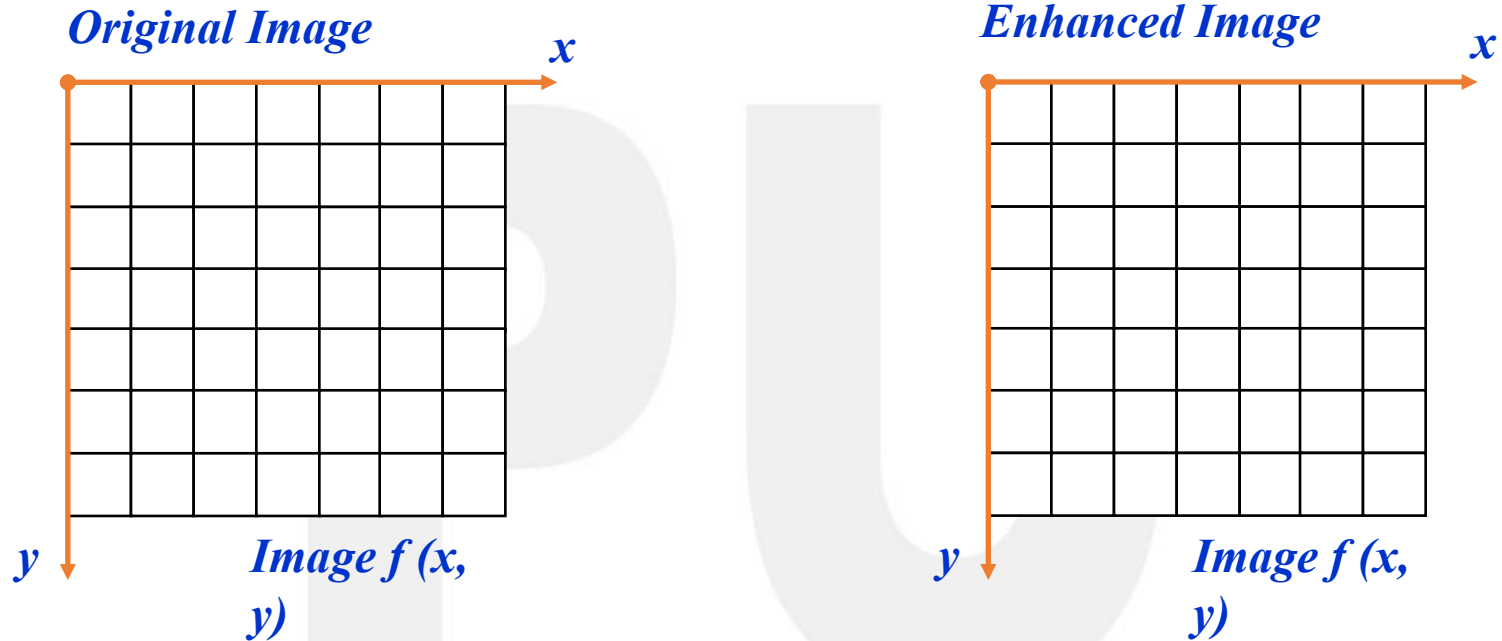
$$s = c * r^\gamma$$

Map a narrow range of dark input values into a wider range of output values or vice versa Varying γ gives a whole family of curves

FIGURE Plots of the gamma equation $s = c r^\gamma$ for various values of γ ($c = 1$ in all cases). Each curve was scaled *independently* so that all curves would fit in the same graph. Our interest here is on the *shapes* of the curves, not on their relative values.



Power Law Transformations (cont...)

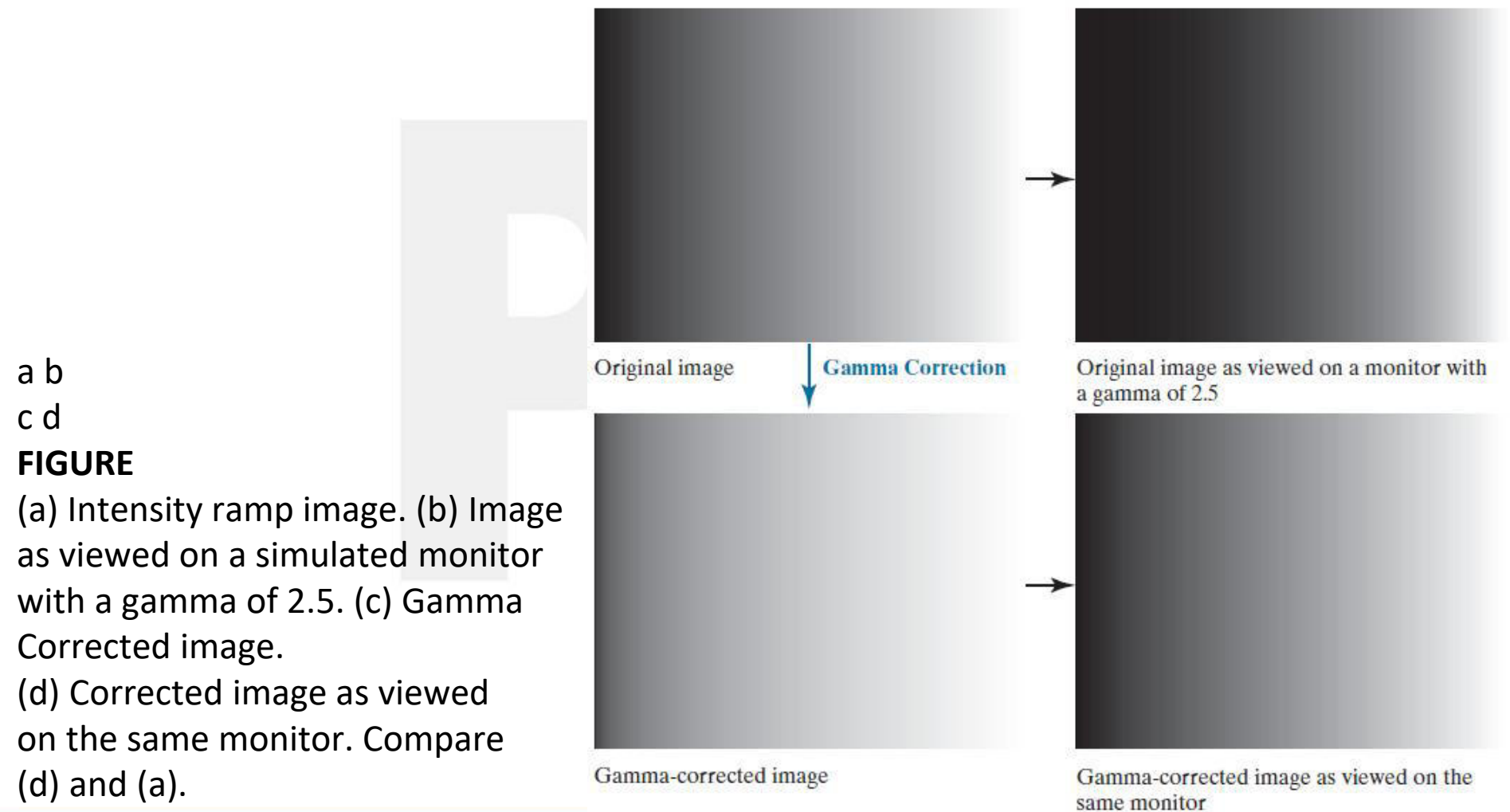


$$s = cr^\gamma$$

We usually set c to 1

Grey levels must be in the range $[0.0, 1.0]$

Cont..



Cont..

a b

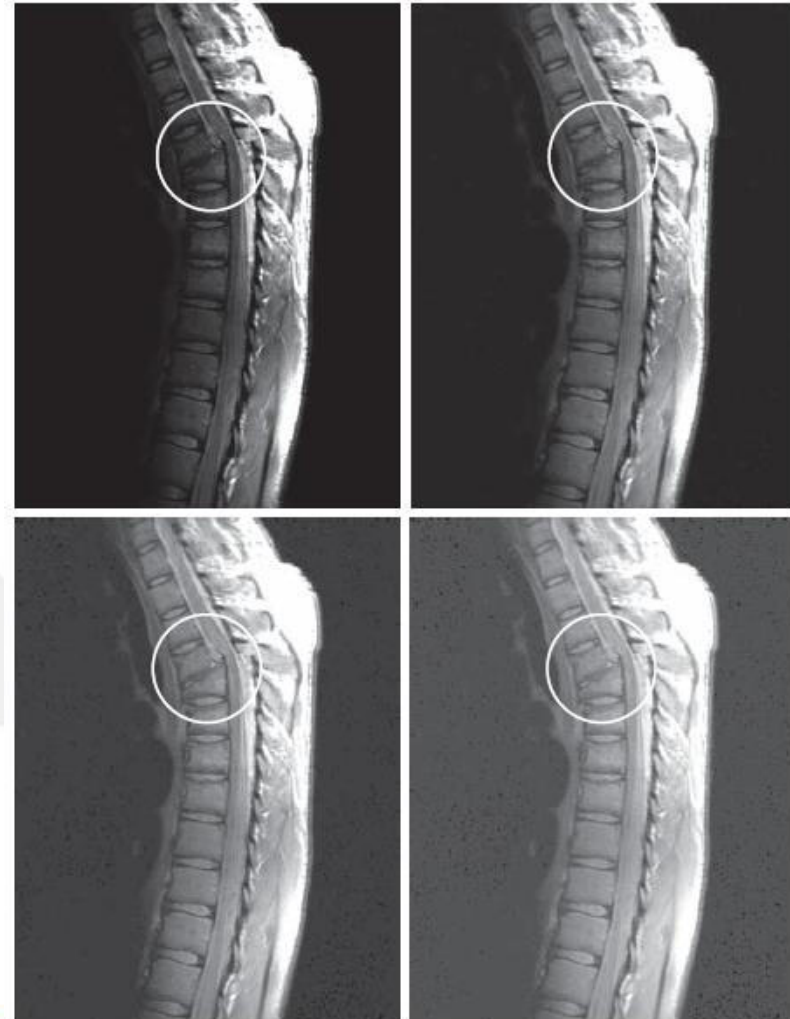
c d

FIGURE

(a) Magnetic resonance image (MRI) of a fractured human spine (the region of the fracture is enclosed by the circle).

(b)–(d) Results of applying the Transformation in Eq. (3-5) with $c = 1$ and $\gamma = 0.6, 0.4,$ and 0.3 , respectively.

(Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)



Power Law Example

P



a b
c d

FIGURE

(a) Aerial image. (b)–(d) Results of applying the transformation in Eq. with $\gamma = 3.0$, 4.0 , and 5.0 , respectively. ($c = 1$ in all cases.) (Original image courtesy of NASA.)

Cont...

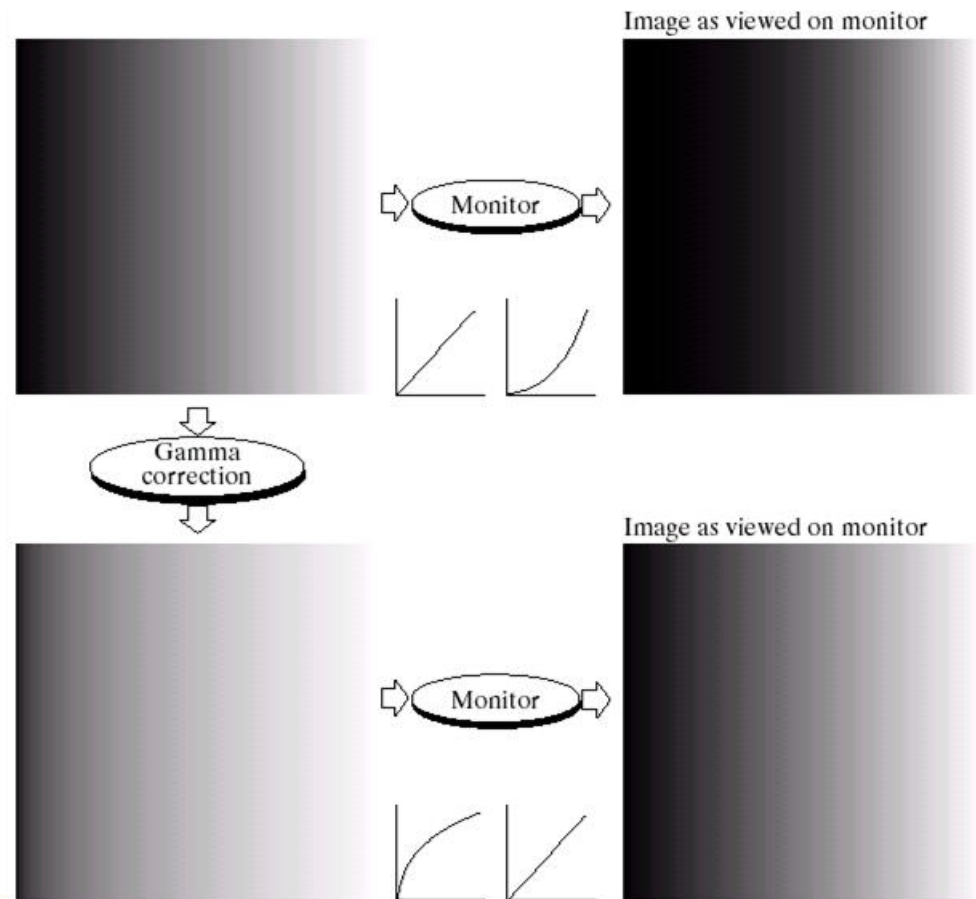
- A variety of devices used for image capture, printing and display respond according to a power law.
- By conventions, the exponent in the power law equation is referred to as gamma.
- The process used to correct these power-law response phenomena is called gamma correction.
- Ex: The CRT devices have an intensity to voltage response that is a power function, with components varying from approximately 1.8-2.5.
- W. R. T. the curve $\gamma=2.5$ display systems would tend to produce image that are the darker than intended.

Gamma Correction

Many of you might be familiar with gamma correction of computer monitors.

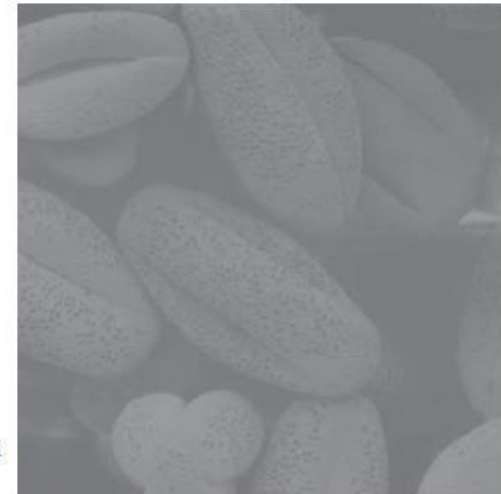
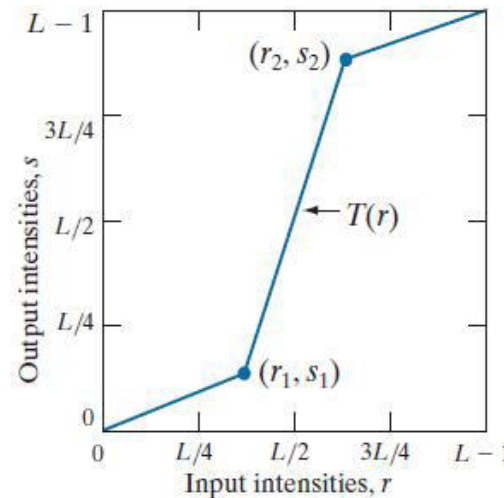
Problem is that display devices do not respond linearly to different intensities

Can be corrected using a log transform



Piecewise Linear Transformation Functions

Rather than using a well defined mathematical function we can use arbitrary user-defined transforms. The images below show a contrast stretching linear transform to add contrast to a poor quality image



a b
c d

FIGURE

Contrast stretching. (a) Piecewise linear Transformation function. (b) A lowcontrast Electron microscope image of pollen, magnified 700 times.

(c) Result of contrast stretching.

(d) Result of thresholding. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)



Cont..

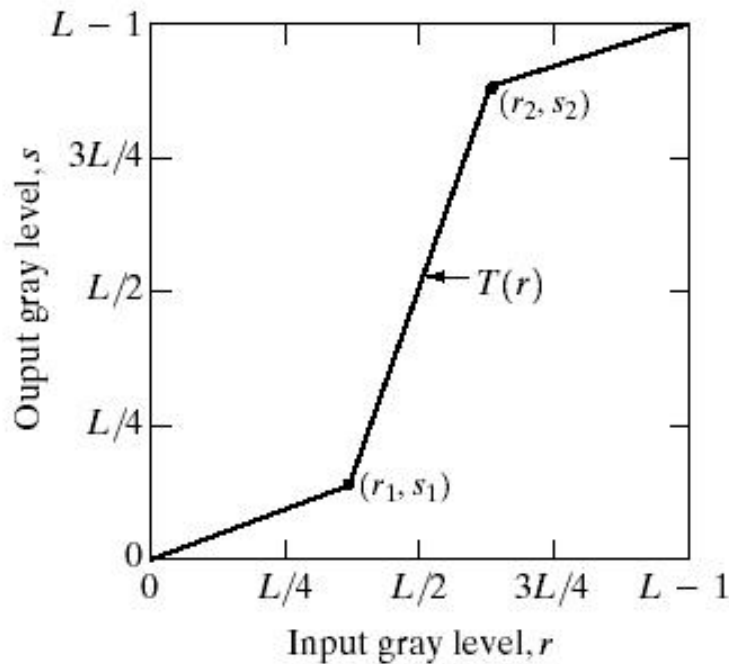
Simplest Piecewise linear function

Low contrast images can result from:

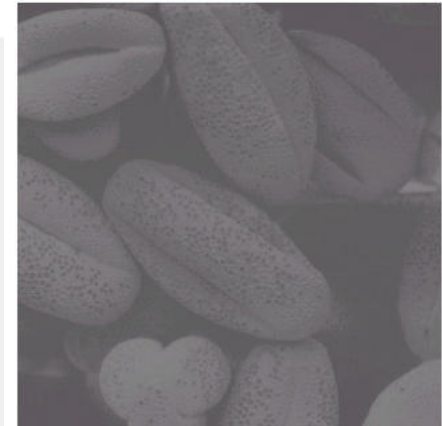
- Poor illumination
- Lack of dynamic range in the imaging sensor
- Wrong setting of a lens aperture during image acquisition
- Contrast stretching is a process that expands the range of intensity levels in an image so that it spans the full intensity range of the recording medium or display device.

Contrast Stretching

Contrast: The difference between the brightest and darkest intensities.



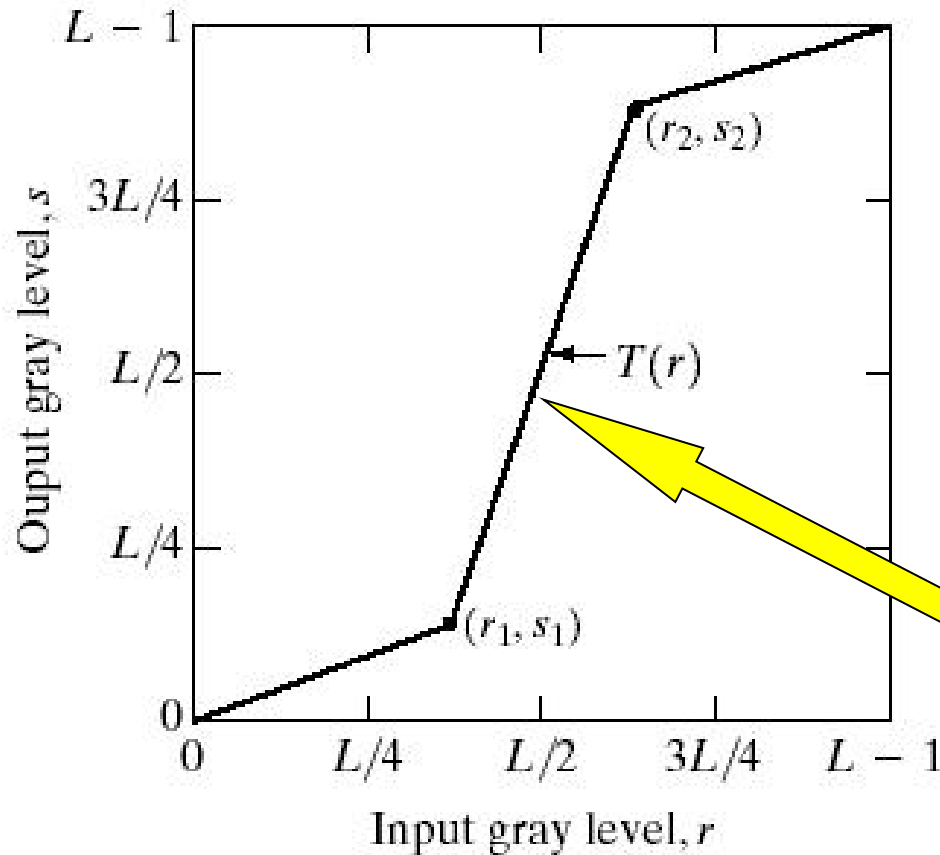
Before contrast enhancement



After contrast enhancement



How to know where the contrast is enhanced ?

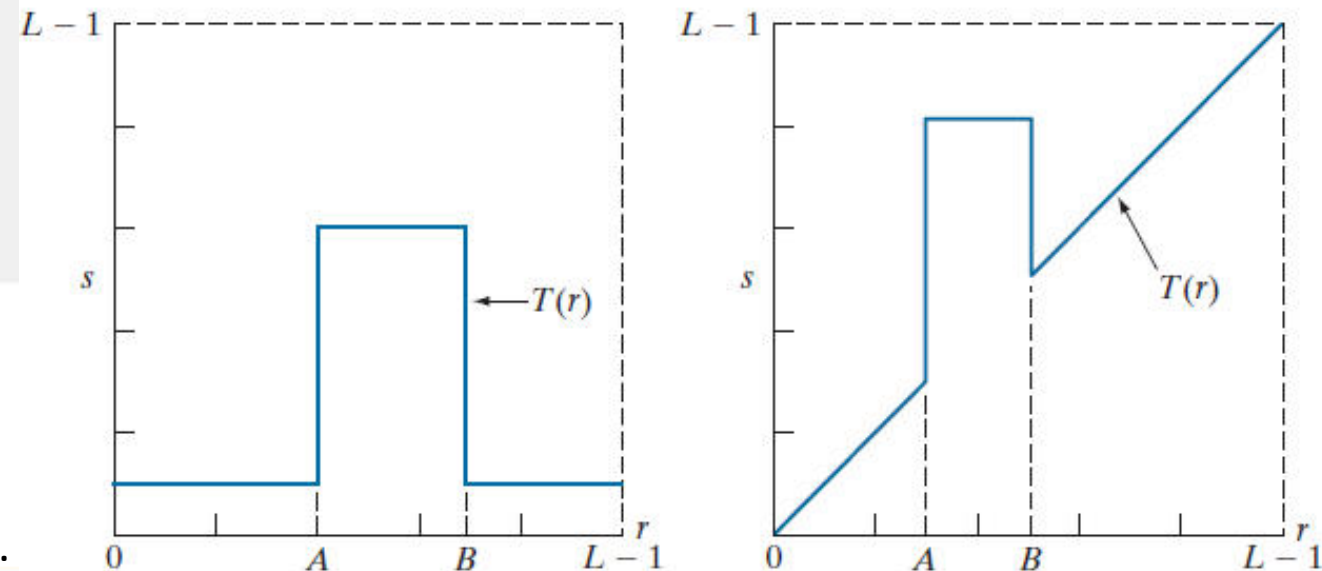


Notice the slope of $T(r)$
 if Slope > 1 ? Contrast increases
 if Slope < 1 ? Contrast decrease
 if Slope $= 1$? no change

Smaller D_r yields wider
 D_s
 = increasing Contrast

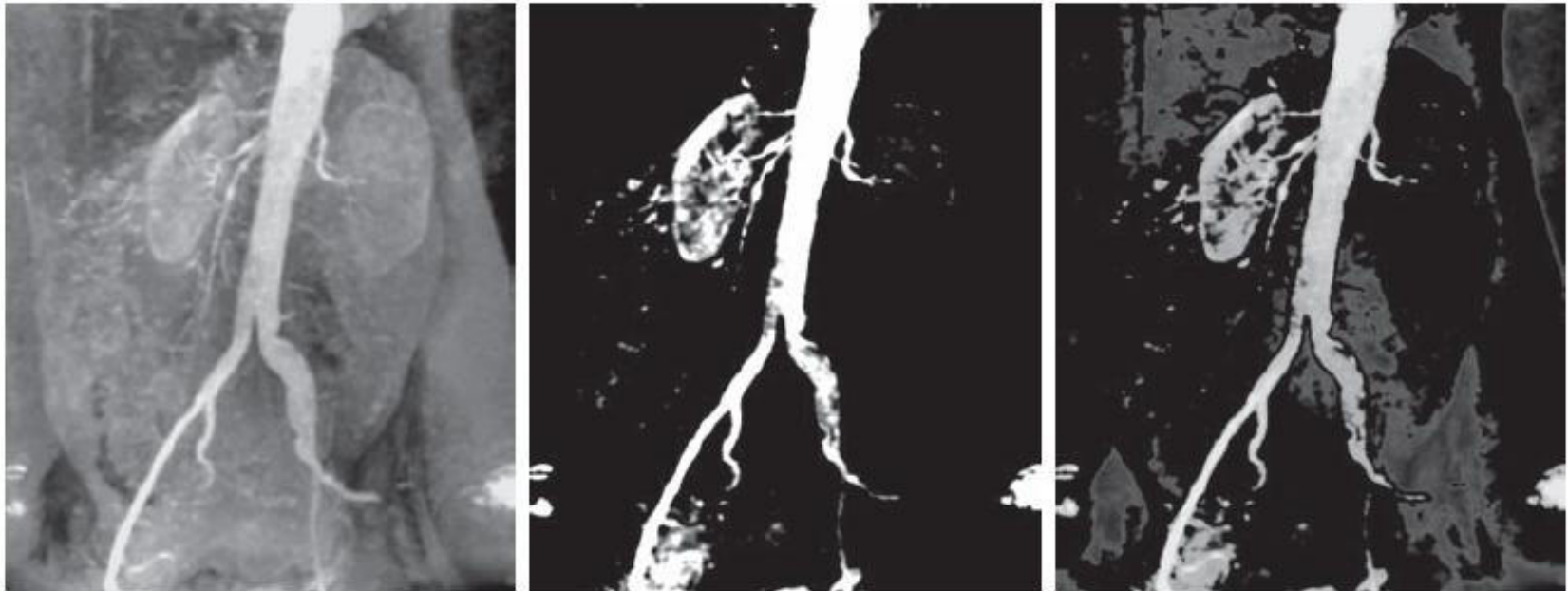
Intensity (Gray) Level Slicing

Highlights a specific range of grey levels: Similar to thresholding, Other levels can be suppressed or maintained, Useful for highlighting features in an image. Applications include: Enhancing feature such as masses of water in satellite imagery, and Enhancing flaws in X-ray Images.



a b
FIGURE
(a) This transformation Function highlights range $[A, B]$ and reduces all other intensities to a lower level.
(b) This function highlights range $[A, B]$ and leaves other intensities unchanged.

Contd..



a b c

FIGURE (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. (a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. (b), with the selected range set near black, so that the grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)

Bit Plane Slicing

Often by isolating particular bits of the pixel values in an image we can highlight interesting aspects of that image

Higher-order bits usually contain most of the significant visual information, and Lower-order bits contain subtle details

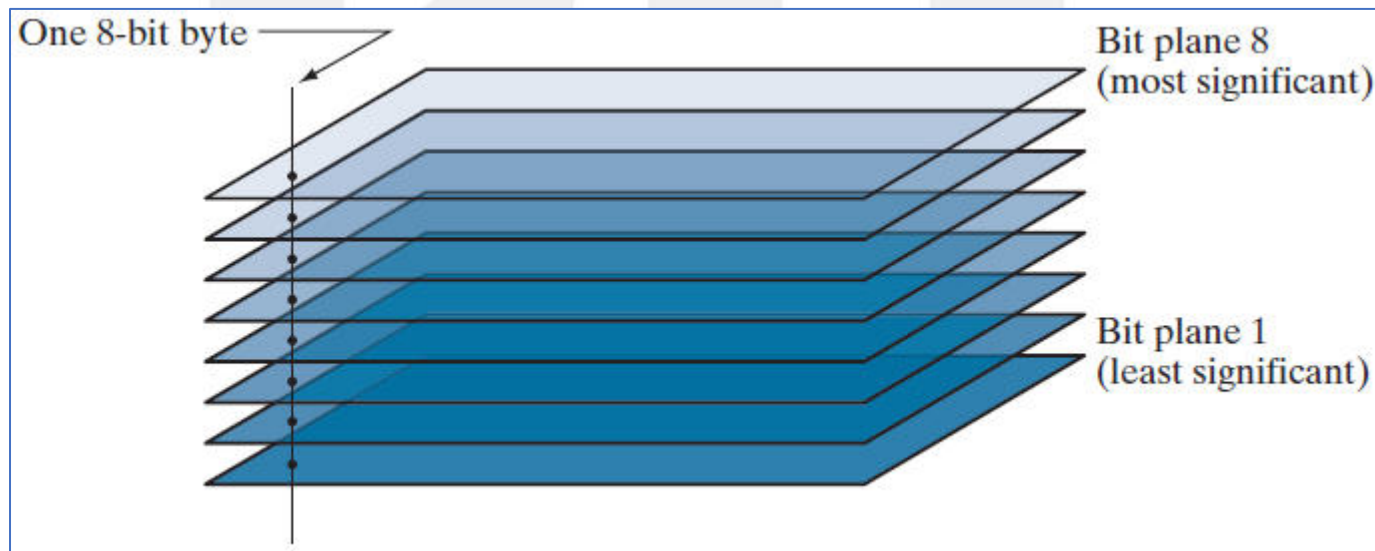
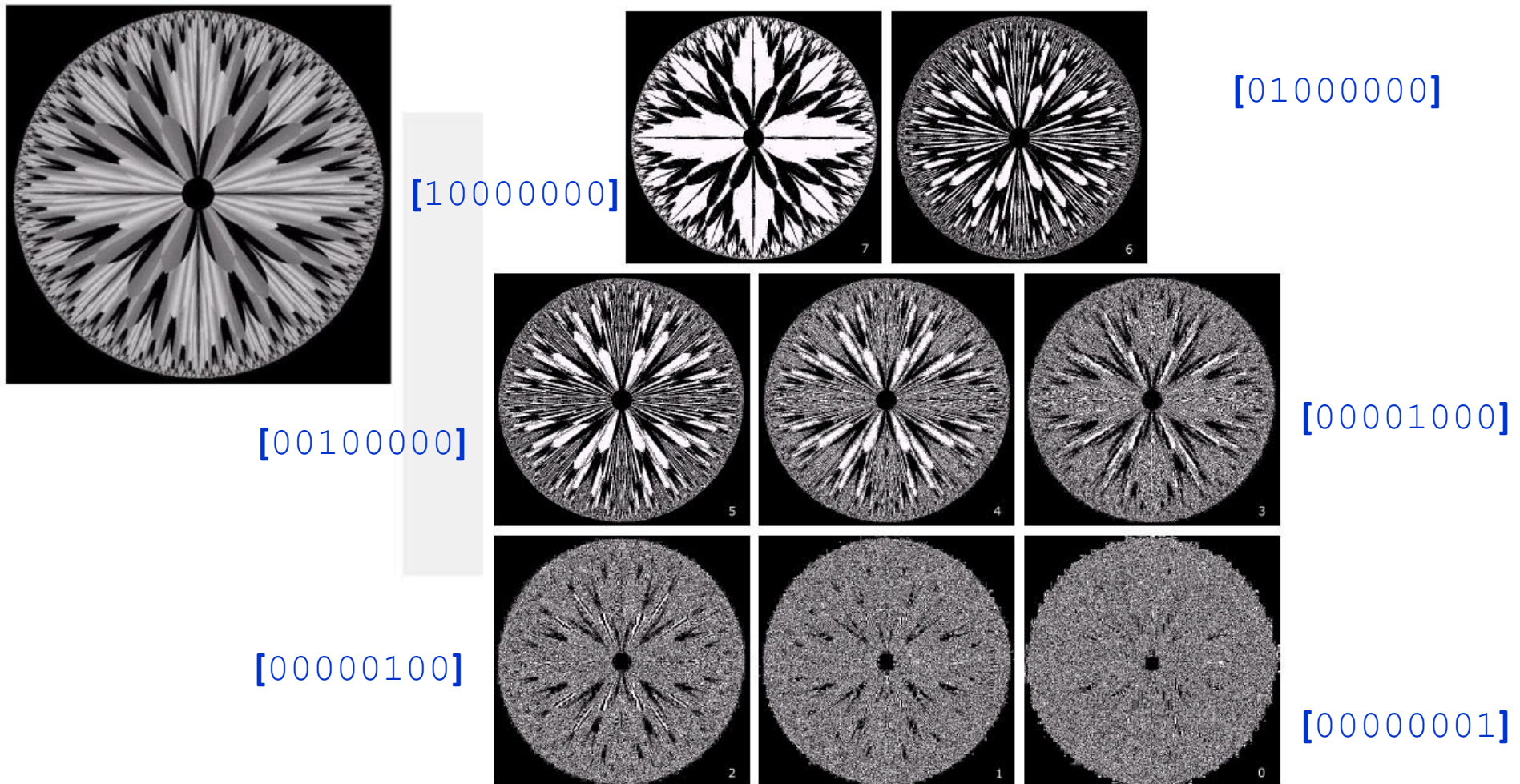
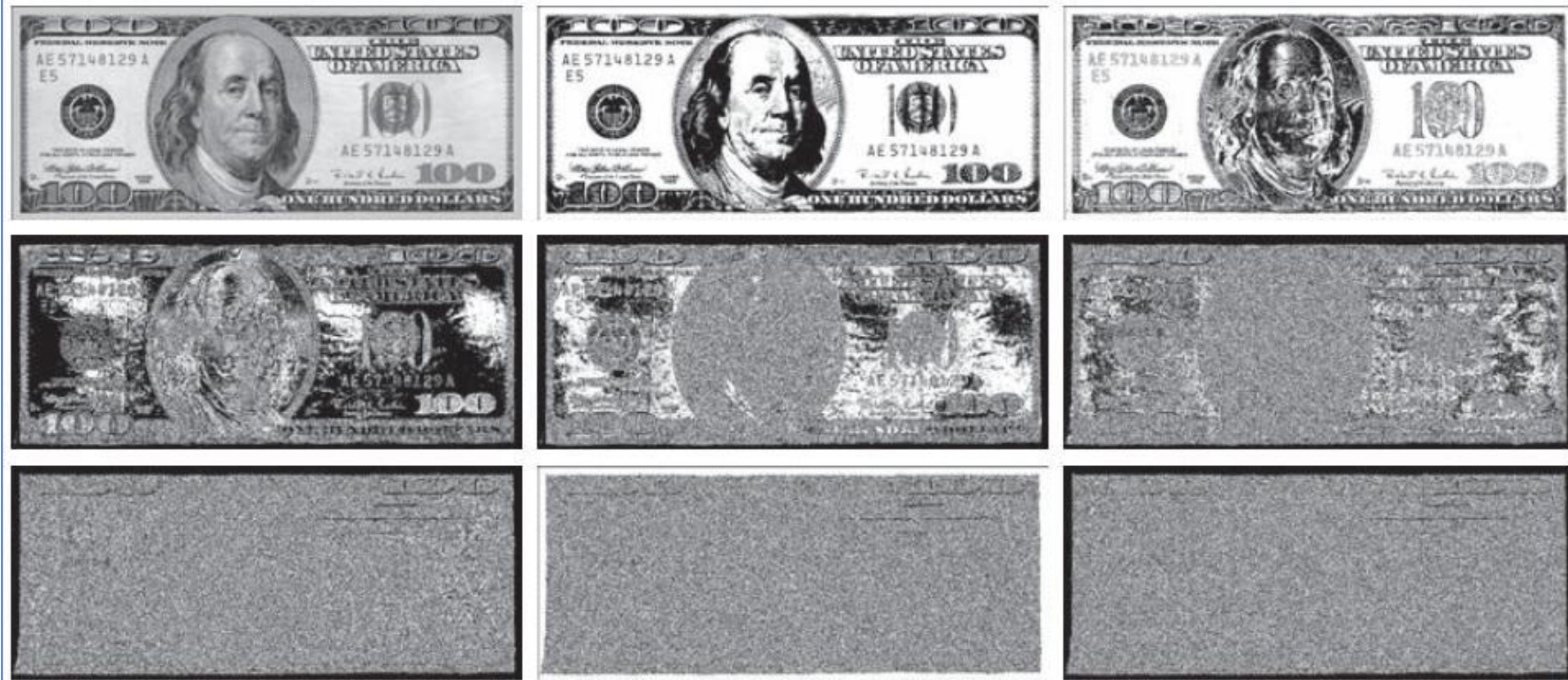


FIGURE Bit-planes of an 8-bit image.

Bit Plane Slicing (cont...)



Bit Plane Slicing (cont...)



a b c
d e f
g h i

FIGURE (a) An 8-bit gray-scale image of size 550×1192 pixels. (b) through (i) Bit planes 8 through 1, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image.

3. Histogram Processing

Histogram = Graph of population frequencies

Grades of the course DIP

- Histogram Equalization
- Histogram Matching
- Local Histogram Processing
- Using Histogram Statistics for Image Enhancement

Histogram Processing

- The histogram of a digital image with intensity levels in the range $[0, L-1]$ is a discrete function. The *unnormalized histogram* of f is defined as
- Where k is the k th intensity value
- n_k is the number of pixels in the image with intensity value k and the subdivisions of the intensity scale are called *histogram bins*.
- Histograms are frequently normalized by the total number of pixels in the image. Assuming a $M \times N$ image, a *normalized histogram*

Cont..

- $P(r_k)$ is related to probability of occurrence of *in the image*
- The sum of all components of normalized histogram is equal to 1.
- Mostly, we work with normalized histograms, which we refer to simply as histograms or image histograms.

Cont..

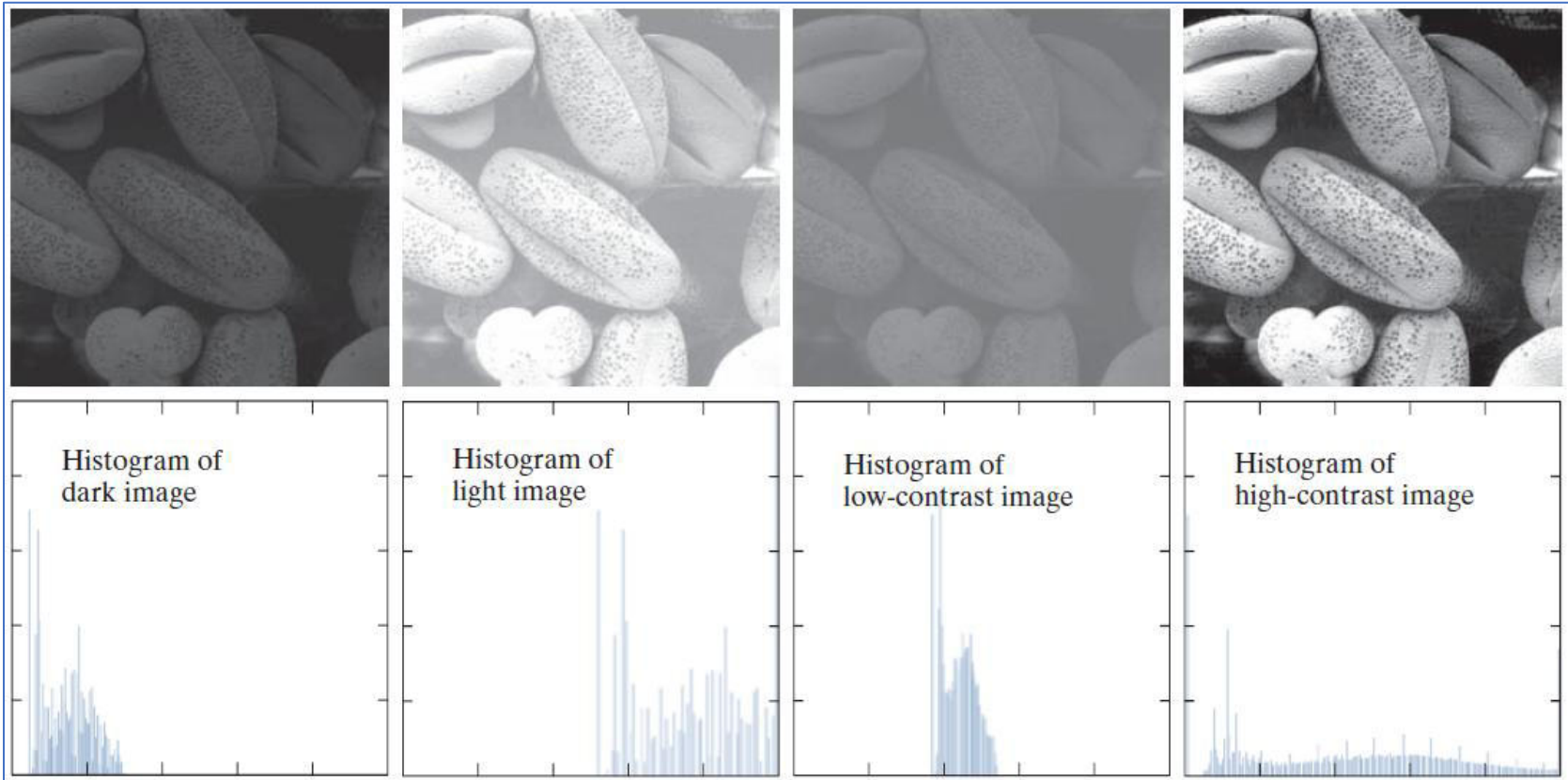


FIGURE a b c d Four image types and their corresponding histograms. (a) dark; (b) light; (c) low contrast; (d) high contrast. The horizontal axis of the histograms are values of r_k and the vertical axis are values of $p(r_k)$.

Cont..

- **Dark Image:** The components of the histogram are concentrated on low (dark) side of the intensity level
- **Light Image:** The components of the histogram are concentrated towards the high (bright) side of the intensity level scale
- **Low contrast image:** Narrow histogram located typically towards the middle of the intensity level.
- **High contrast image:** Covers a wide range of intensity level and further the distribution of pixels is not far from the uniform, with very few vertical lines being much higher than others.

Histogram Equalization

Histogram equalization is a method which increases the dynamic range of the gray-levels in a low-contrast image to cover full range of gray-levels, and its purpose is

- To improve the contrast of an image
- To transform an image in such a way that the transformed image has a nearly **uniform distribution** of pixel values

Transformation:

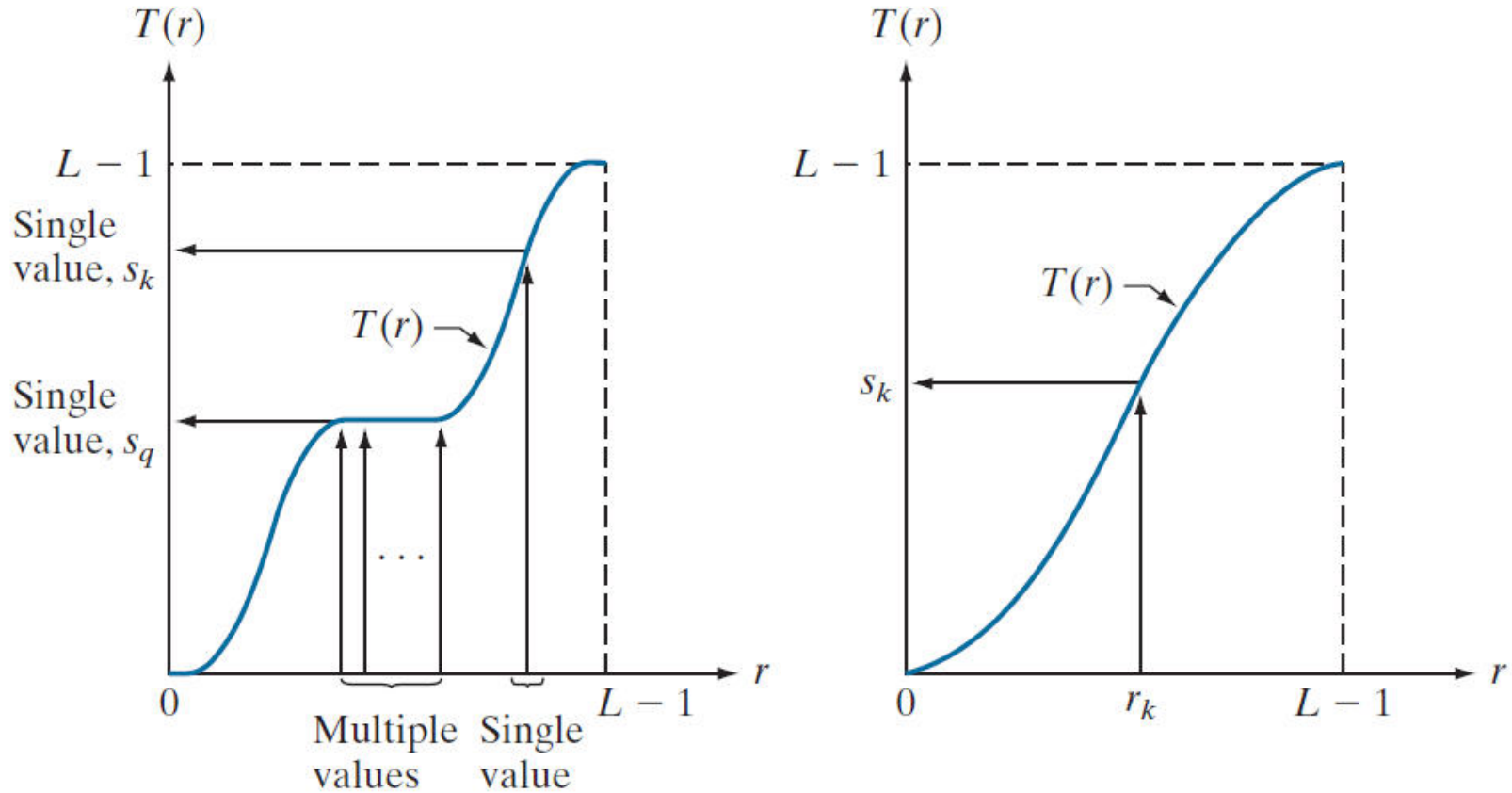
- Assume r has been normalized to the interval $[0,1]$, with $r = 0$ representing black and $r = 1$ representing white

$$s = T(r) \quad 0 \leq r \leq 1$$

Contd..

- The transformation function satisfies the following conditions:
- $T(r)$ is single-valued and monotonically increasing in the interval to prevent reversal intensity artifacts;
- $T(r)$ is strictly monotonically increasing function being in $[0, L-1]$ for r in $[0, L-1]$; to ensure the range of output intensity being the same as for input intensity.

Cont..



a b

FIGURE (a) Monotonic increasing function, showing how multiple values can map to a single value. (b) Strictly monotonic increasing function. This is a one-to-one mapping, both ways.

Histogram Equalization

PU

Cont..

- The condition in (a) that $T(r)$ be monotonically increasing guarantees that output intensity values will never be less than corresponding input values, thus preventing artifacts created by reversals of intensity.
- Condition (b) guarantees that the range of output intensities is the same as the input.
- Finally, condition (a') guarantees that the mappings from s back to r will be one-to-one, thus preventing ambiguities.

Probability Density Function

- The image intensity levels can be viewed as random variables in interval $[0, L-1]$.
- Histogram is analogous to Probability Density Function (PDF) which represent density of population.
- Let s and r be Random variables with PDF $p_s(s)$ and $p_r(r)$ respectively and relation between s and r is

$$s = T(r)$$

- Let , represent PDFs of r and s , respectively. The desired transformation is

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right|$$

Cont..

- Let ,
- Then,

where w is a dummy variable of integration. The integral on the right side is the *cumulative distribution function* (CDF) of random variable r .

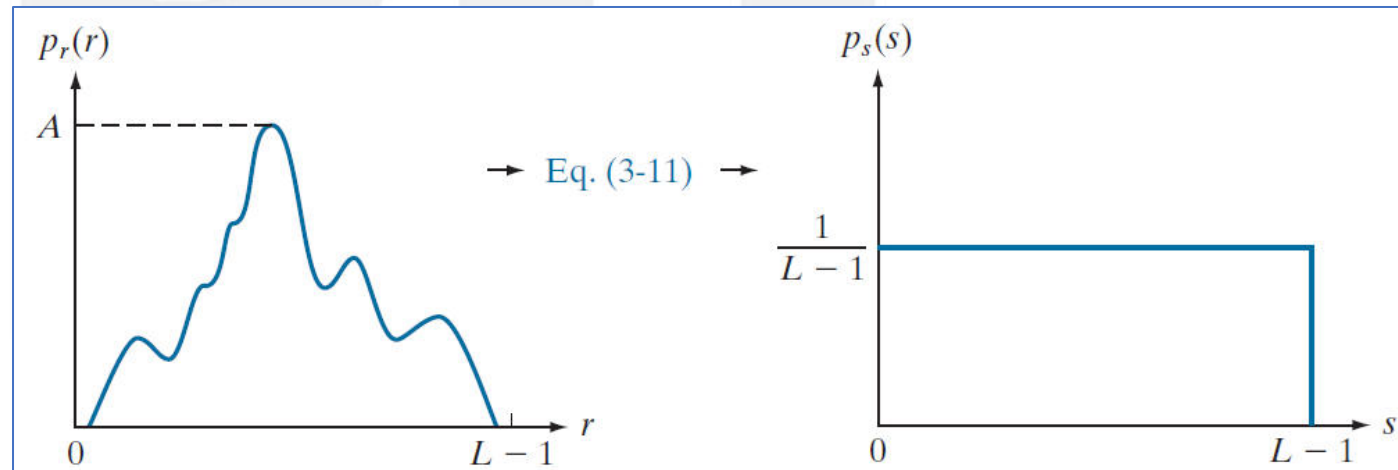
PU

Histogram Equalization

PU

Cont..

- $p_s(s)$ now is a uniform probability density function. $T(r)$ depends on $p_r(r)$, but the resulting $p_s(s)$ always is uniform. The probability function of the output levels is uniform, that is: The transformation generates an image whose intensity levels are equally likely covering the entire range $[0,1]$. The net result is the intensity-level equalization. Note that the transformation function is simply the cumulative distribution function (CDF).



a b

FIGURE (a) An arbitrary PDF. (b) Result of applying Eq. to the input PDF. The resulting PDF is always uniform, independently of the shape of the input.

Example

PU

Example

PU

Example

PU

Cont..

In discrete version: The probability of occurrence of gray level r_k in an image is

MN : Total number of pixels in the image

n_k : Number of pixels that have gray level r_k

L : Total number of possible gray levels in the image

❑ The transformation function is

❑ Thus, an output image is obtained by mapping each pixel with level r_k in the input image into a corresponding pixel with level s_k . The transformation $T(r_k)$ in this equation is called Histogram Equalization or Histogram Linearization transformation.

Cont..

This will not produce a uniform histogram, but will tend to spread the histogram of the input image

Advantages:

- Gray-level values cover entire scale (contrast enhancement)
- Fully automatic

Cont..

•

PU

Cont..

Suppose that a 3-bit image ($L=8$) of size 64×64 pixels ($MN = 4096$) has the intensity distribution shown in following table. Get the histogram equalization transformation function and give the $p_s(s_k)$ for each s_k .

r_k	n_k	$p_r(r_k) = n_k / MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

Cont..

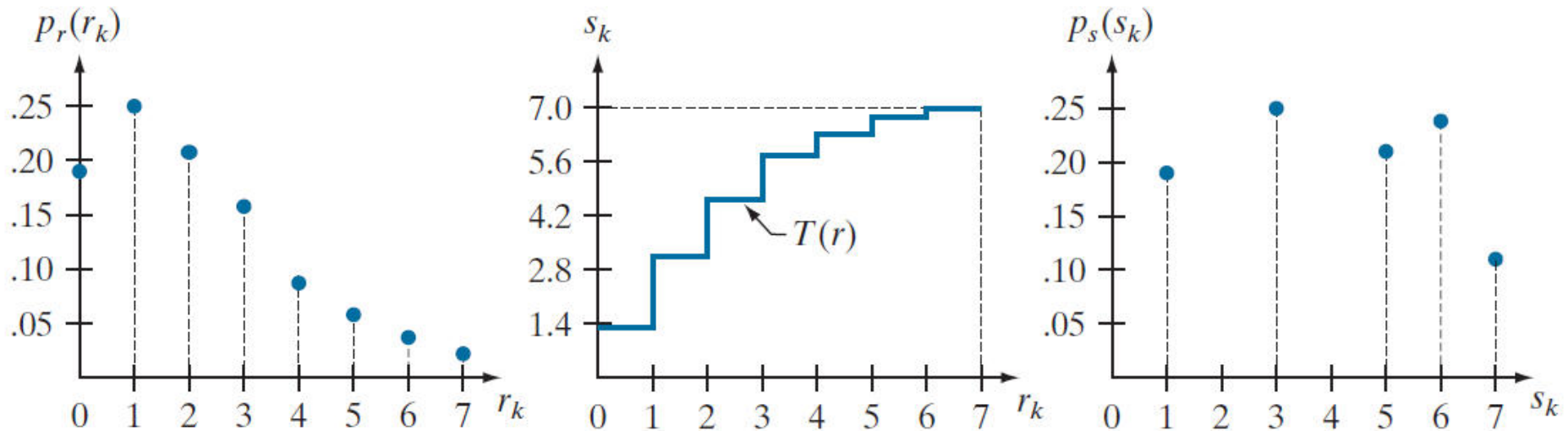
r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

Cont..

These are the values of the equalized histogram. You can see now there only five gray levels.

r_k	n_k	s_k	<i>New n_k</i>	$p_s(s_k) = \text{New } n_k / MN$
$r_0 = 0$	790	$s_0 = 1$	790	0.19
$r_1 = 1$	1023	$s_1 = 3$	1023	0.25
$r_2 = 2$	850	$s_2 = 5$	850	0.21
$r_3 = 3$	656	$s_3 = 6$	985	0.24
$r_4 = 4$	329	$s_4 = 6$		
$r_5 = 5$	245	$s_5 = 7$	448	0.11
$r_6 = 6$	122	$s_6 = 7$		
$r_7 = 7$	81	$s_7 = 7$		

Cont..



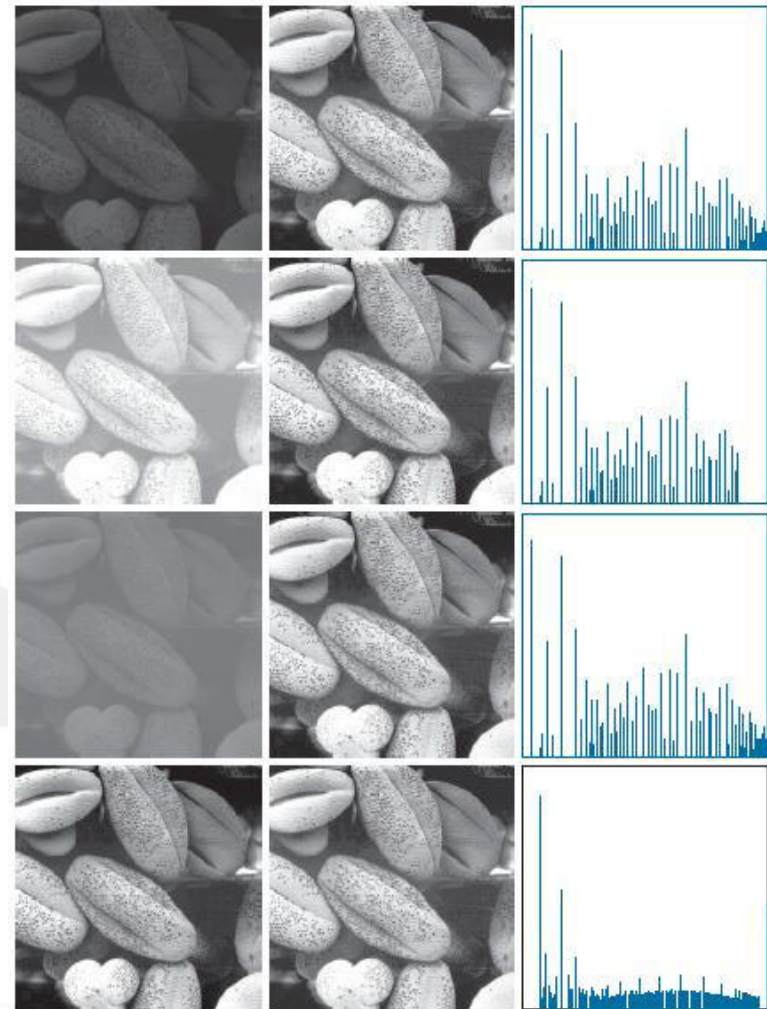
$$\begin{array}{llll} s_0 = 1.33 \rightarrow 1 & s_2 = 4.55 \rightarrow 5 & s_4 = 6.23 \rightarrow 6 & s_6 = 6.86 \rightarrow 7 \\ s_1 = 3.08 \rightarrow 3 & s_3 = 5.67 \rightarrow 6 & s_5 = 6.65 \rightarrow 7 & s_7 = 7.00 \rightarrow 7 \end{array}$$

a b c

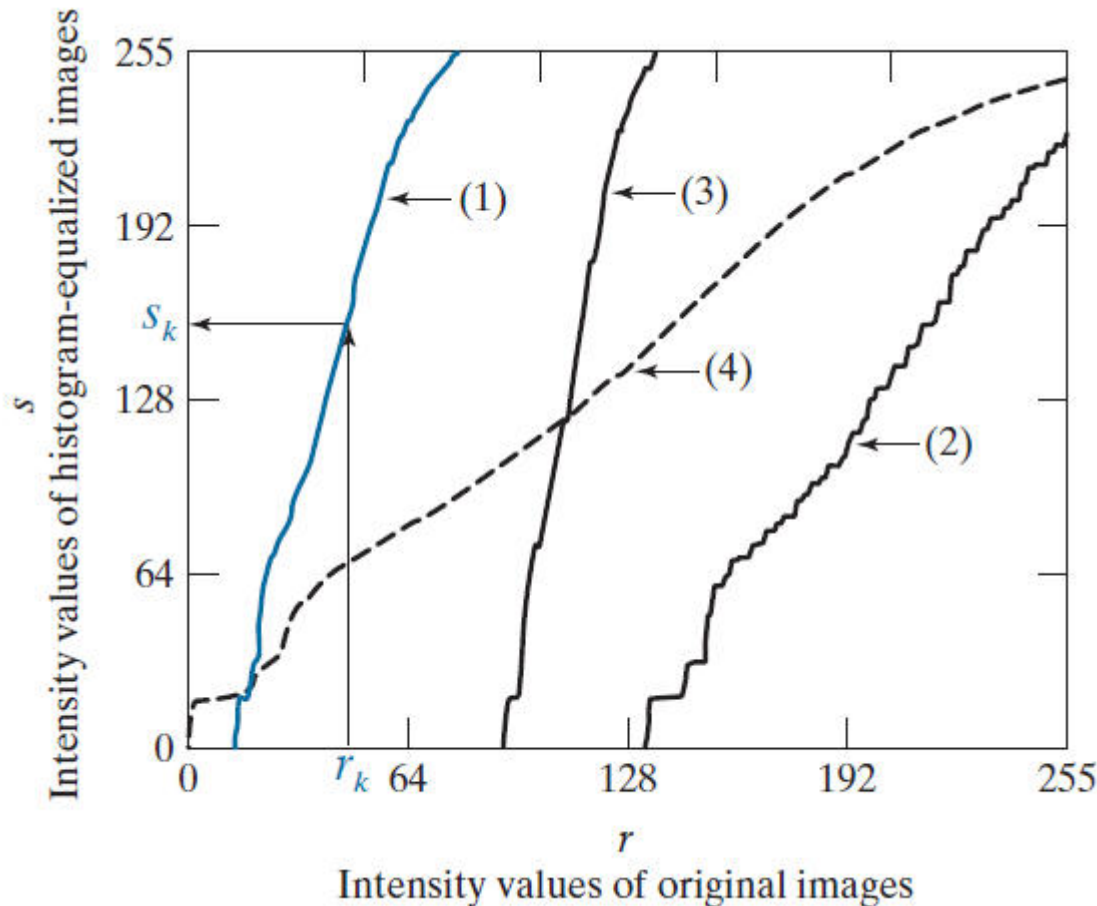
FIGURE Histogram equalization. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.

Cont..

FIGURE Left column: Original Images .
Center column: Corresponding histogram-
equalized images.
Right column: histograms of the images in the
center column.



Cont..



FIGURE

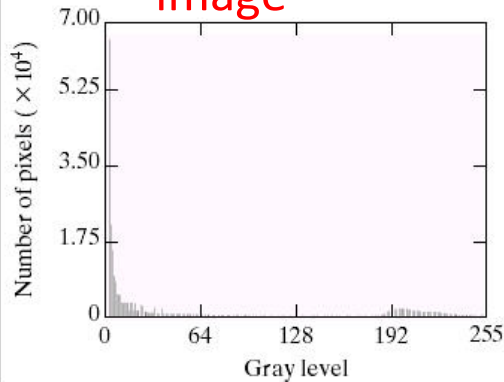
Transformation functions for histogram equalization. Transformations (1) through (4) were obtained using Eq. and the histograms of the images on the left column. Mapping of one intensity value r_k in image 1 to its corresponding value s_k is shown.

Is histogram equalization always good? NO

Histogram Equalization (cont.)



Original
image

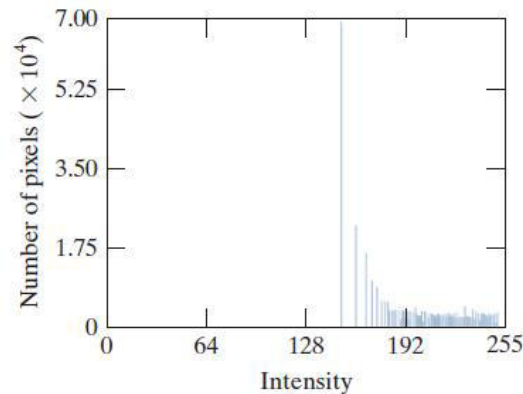
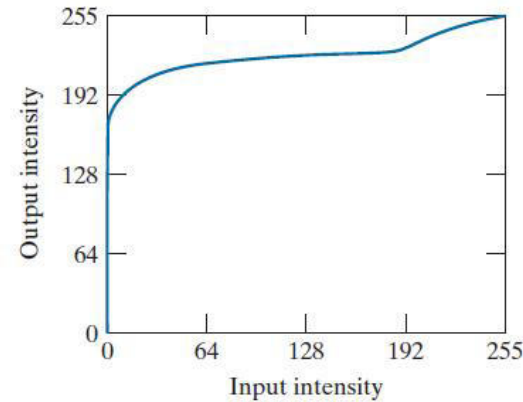


a b

FIGURE

(a) An image, and (b) its histogram.

After histogram equalization, the image become a low contrast image



a b

c

FIGURE

(a) Histogram equalization transformation obtained using the histogram. (b) Histogram equalized image. (c) Histogram of equalized image.

Histogram Matching (histogram specification)

Generate a processed image that has a specified histogram

PU

Histogram Matching

PU

- Obtain $p_r(r)$ from the input image and then obtain the values of s
- Use the specified PDF and obtain the transformation function $G(z)$
- Mapping from s to z

Cont..

Obtain the scaled histogram-equalized values,

Compute all the values of the transformation function G ,

a b

c d

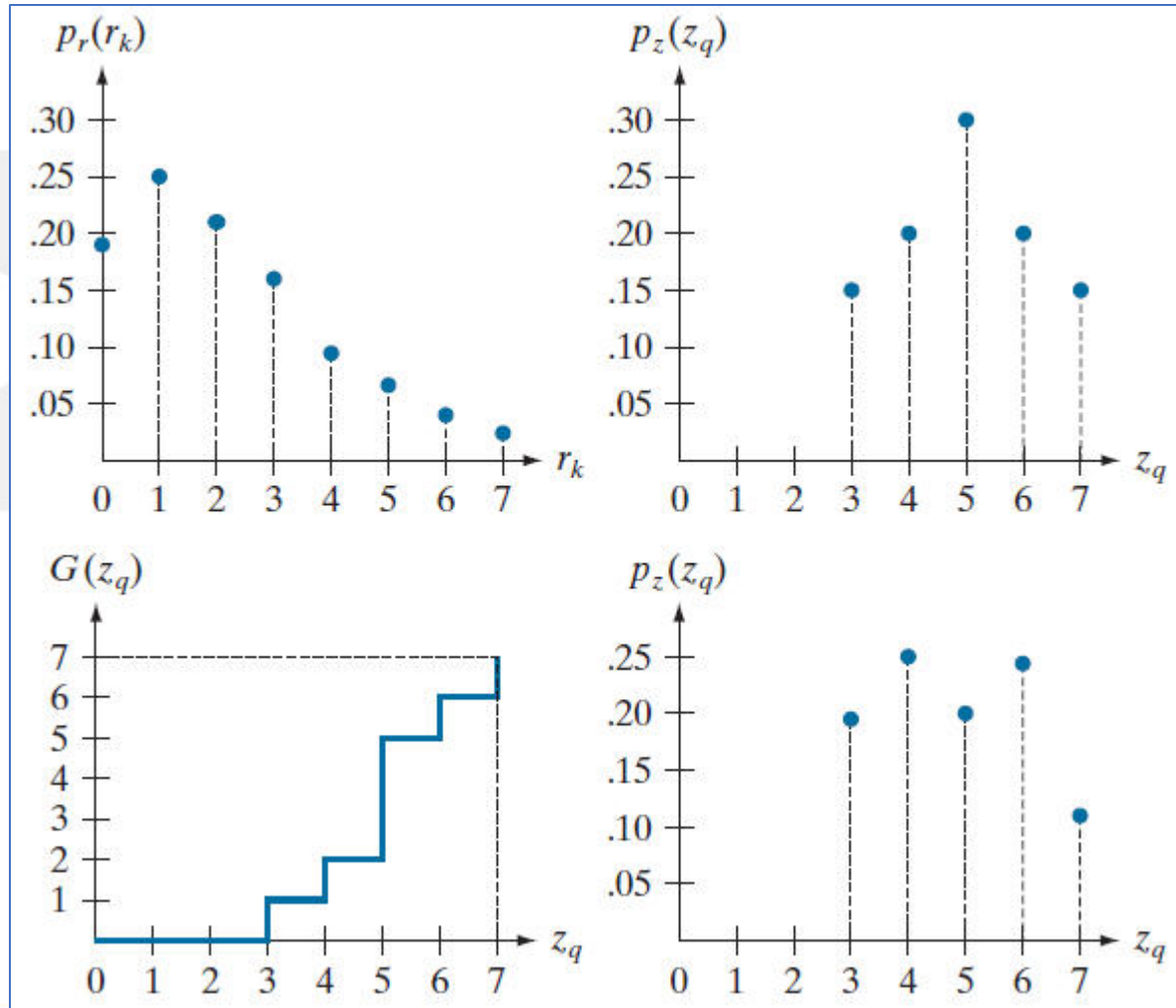
FIGURE

(a) Histogram of a 3-bit image. (b) Specified histogram.

(c) Transformation function obtained from the specified histogram.

(d) Result of histogram specification.

Compare the histograms in (b) and (d).



Cont..

Obtain the scaled histogram-equalized values,

Compute all the values of the transformation function G ,

s_0

s_1

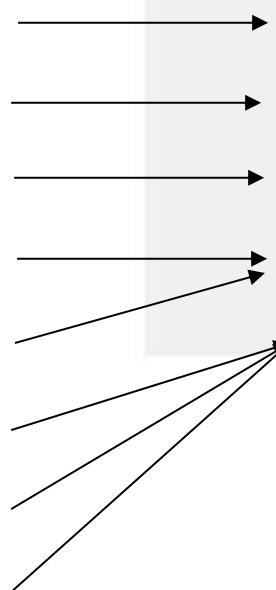
s_2

s_3 s_5

s_4 s_6 s_7

Example: Histogram Matching

s_k	$\rightarrow$	z_q
1	$\rightarrow$	3
3	$\rightarrow$	4
5	$\rightarrow$	5
6	$\rightarrow$	6
7	$\rightarrow$	7



Example: Histogram Matching

a c
b
d

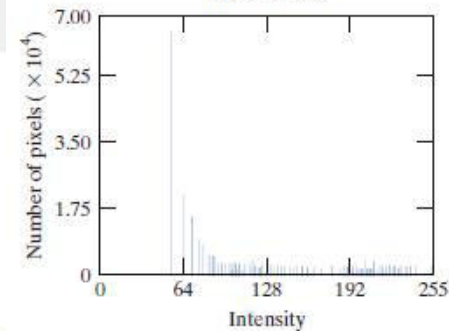
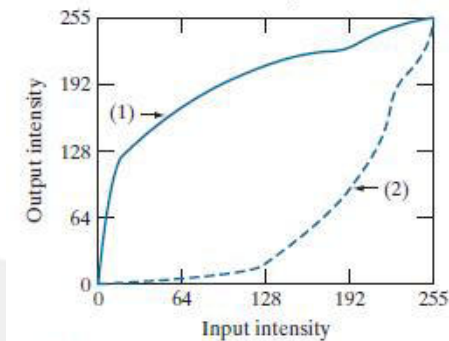
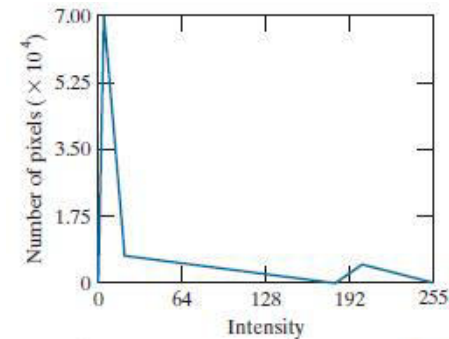
FIGURE

Histogram specification.
(a) Specified histogram.
(b) Transformation $G(zq)$, labeled (1), and $G^{-1}(sk)$, labeled (2).
(c) Result of Histogram specification.
(d) Histogram of image (c).

Desired
histogram

Transfer
function

Actual
histogram



Cont..

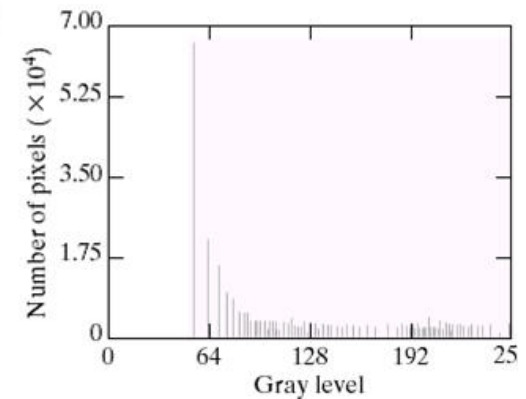
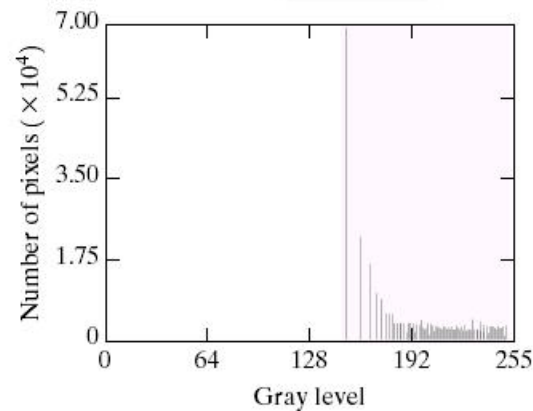
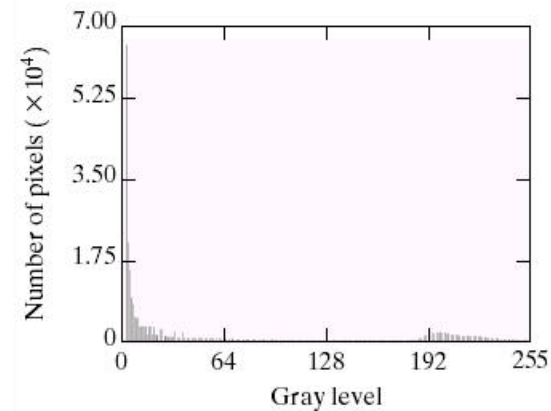
Original
image



After
histogram
equalization



After
histogram
matching



Local Histogram Processing

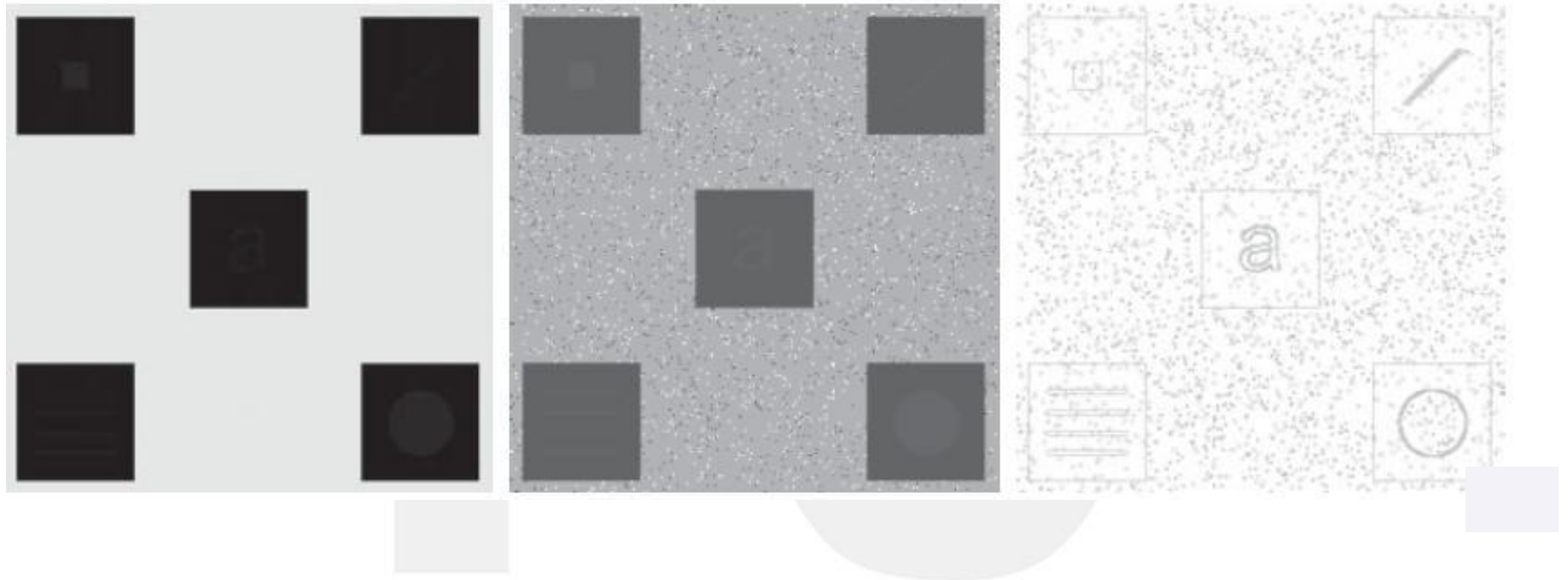
Define square or rectangular neighbourhood (mask) and move the center from pixel to pixel

For each neighbourhood...

- Calculate histogram of the points in the neighbourhood
- Obtain histogram equalization/specification function
- Map gray level of pixel centered in neighbourhood
- Can use new pixel values and previous histogram to calculate next histogram

Move to the next location and repeat the procedure

Local Histogram Processing: Example



a b c

FIGURE

(a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization.

Using Histogram Statistics for Image Enhancement

Average Intensity

The mean value of r :

The n th moment of r (discrete) about its mean is defined as

Note that $\mu_0 = 1$ and $\mu_1 = 0$, and that μ_2 is the variance

Variance

Mean: measure of average intensity (gray level)

Variance (or standard deviation): measure of average contrast in an image

Cont..

The mean and variance can be estimated directly from the samples values, without computing the histogram.

Appropriately these estimates are known as samples mean and sample variance, and given by following familiar expressions from basic statistics:

PU

The global mean and variance are computed over an entire image and are useful for gross adjustments in overall intensity and contrast

Cont..

Two uses of the **mean and variance** for enhancement purposes:

- The **global** mean and variance (global means for the entire image) are useful for adjusting overall contrast and intensity.
- The mean and standard deviation for a **local** region are useful for correcting for large-scale changes in intensity and contrast.

Local Enhancement : Histogram Statistic for Image Enhancement

We can use statistic parameters such as Mean, Variance of Local area for image enhancement

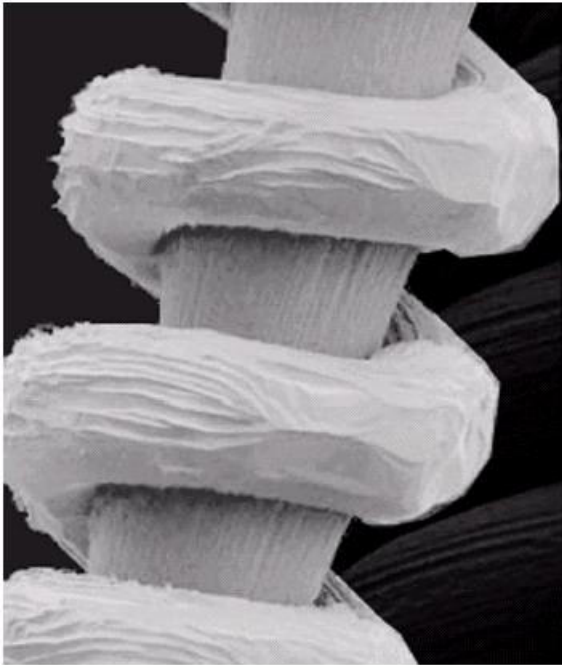


Image of tungsten filament taken using An electron microscope

In the lower right corner, there is a filament in the background which is very dark and we want this to be brighter.

We cannot increase the brightness of the whole image since the white filament will be too bright.

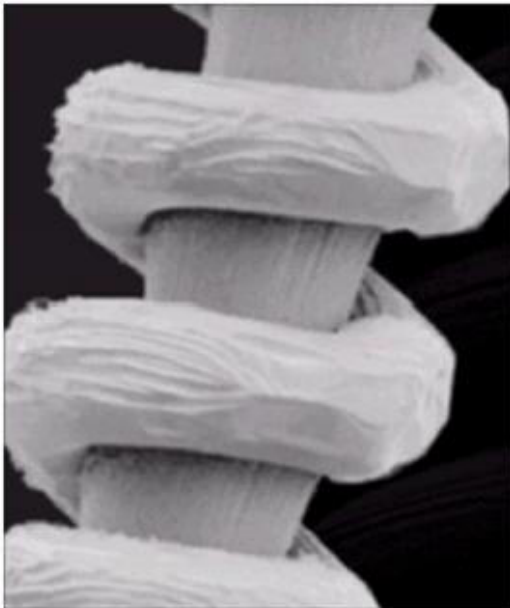
Contd...

PU

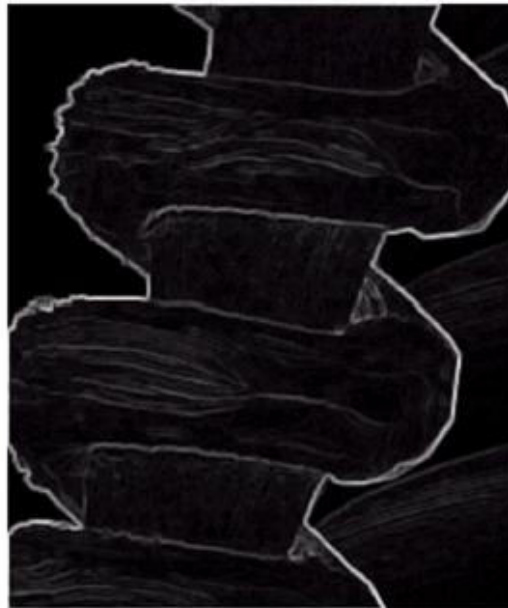
Example: Local enhancement for this task

$$g(x, y) = \begin{cases} E \cdot f(x, y) & \text{when } m_{s_{xy}} \leq k_0 M_G \text{ and } k_1 D_G \leq \sigma_{s_{xy}} \leq k_2 M_G \\ f(x, y) & \text{otherwise} \end{cases}$$

Original image



Local Variance
image

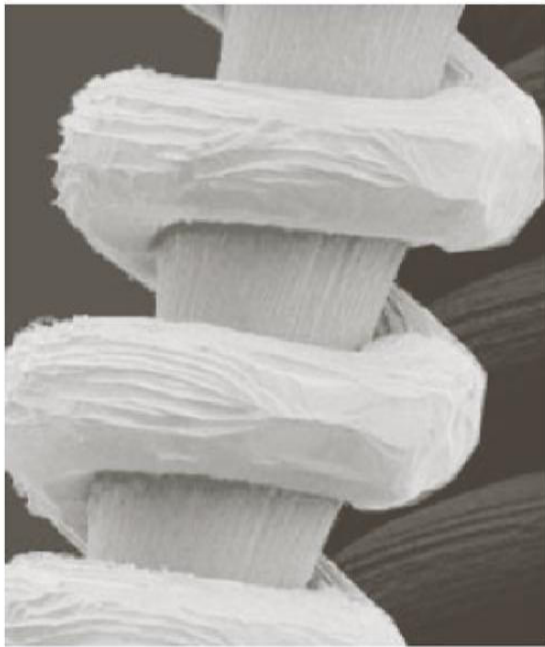


Multiplication
factor



Using Histogram Statistics for Image Enhancement: Example

Original SEM Image



Global Histogram
Equalization Image

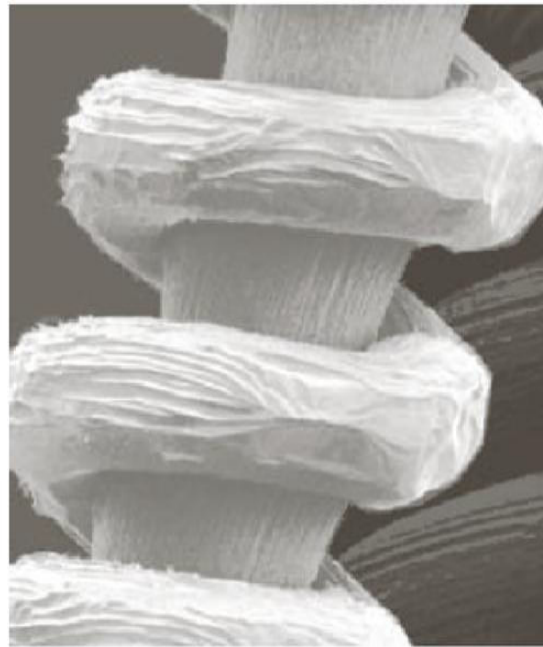


Image Enhanced using
local histogram statistics



Spatial Filtering

A spatial filter consists of (a) a **neighborhood**, and (b) a **predefined operation**

Linear spatial filtering of an image of size $M \times N$ with a filter of size $m \times n$ is given by the expression

5.Smoothing Spatial Filters

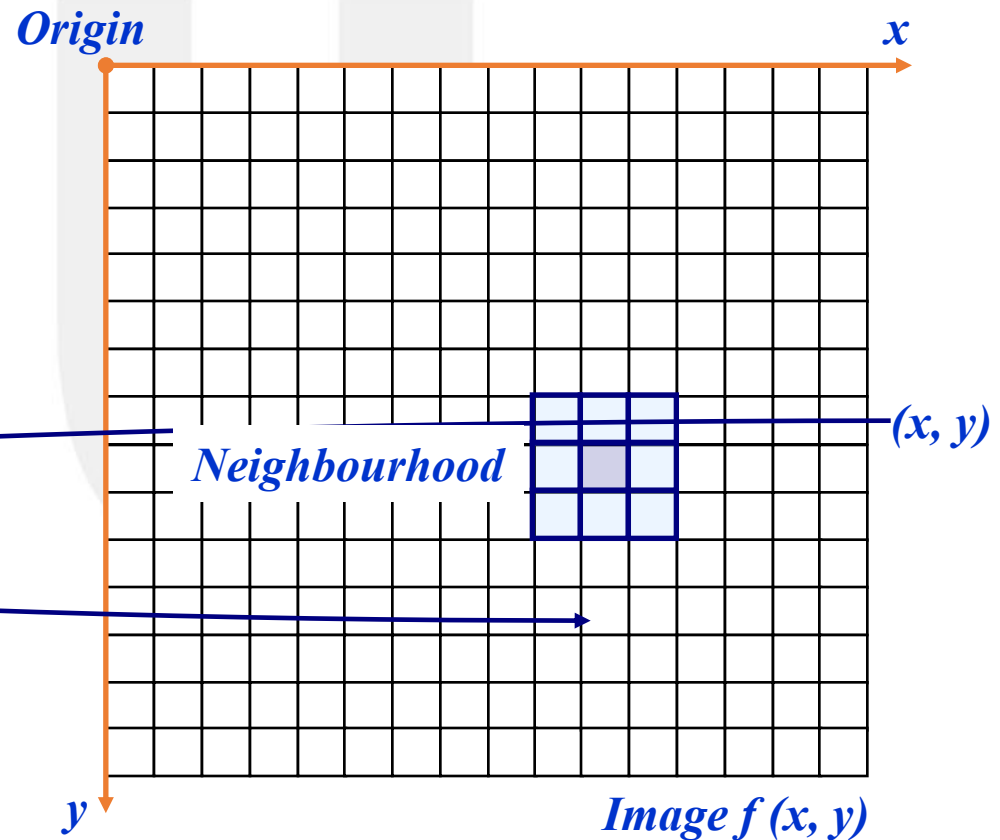
- Smoothing filters are used for blurring and for noise reduction
- Blurring is used in removal of small details and bridging of small gaps in lines or curves
- Smoothing spatial filters include linear filters and nonlinear filters.

Neighbourhood Operations

Neighbourhood operations simply operate on a larger neighbourhood of pixels than point operations

Neighbourhoods are mostly a rectangle around a central pixel

Any size rectangle and any shape filter are possible



Simple Neighbourhood Operations

Some simple neighbourhood operations include:

- **Min:** Set the pixel value to the minimum in the neighbourhood
- **Max:** Set the pixel value to the maximum in the neighbourhood
- **Median:** The median value of a set of numbers is the midpoint value in that set (e.g. from the set [1, 7, 15, 18, 24] 15 is the median). Sometimes the median works better than the average

Simple Neighbourhood Operations Example

Original Image

123	127	128	119	115	130	
140	145	148	153	167	172	
133	154	183	192	194	191	• • •
194	199	207	210	198	195	
164	170	175	162	173	151	
						• • •

x

y

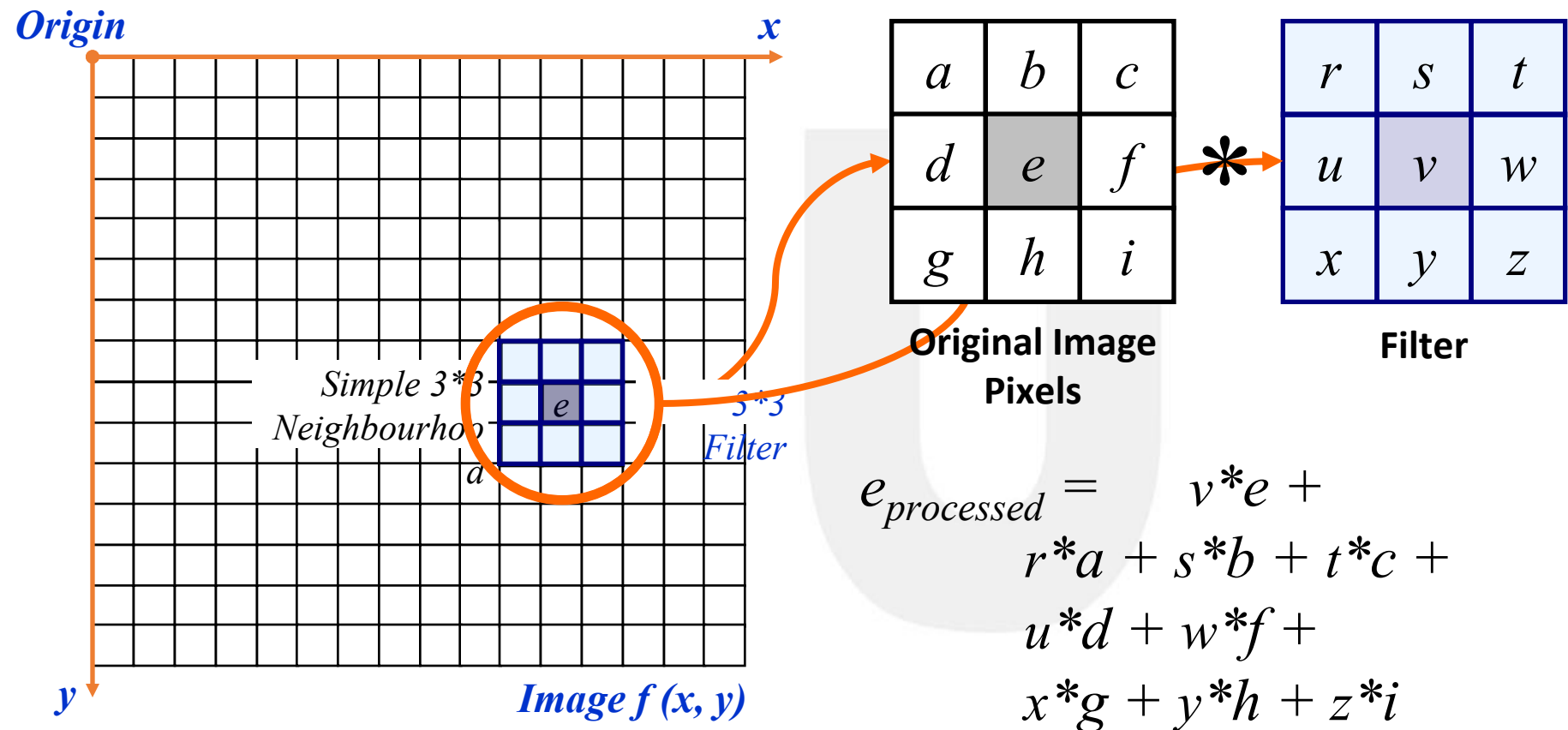
Enhanced Image

						• • •
						• • •

x

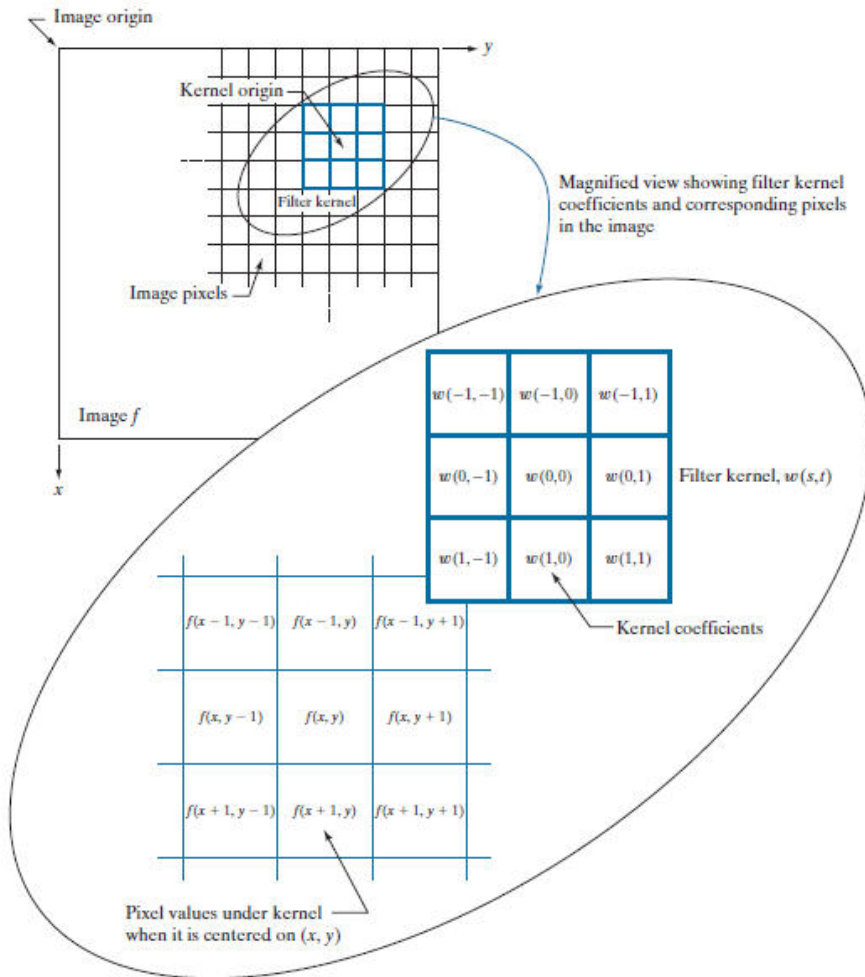
y

The Spatial Filtering Process



The above is repeated for every pixel in the original image to generate the filtered image

Spatial Filtering: Equation Form



$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x+s, y+t)$$

Filtering can be given in equation form as shown above. Notations are based on the image shown to the left.

Smoothing Spatial Filters

One of the simplest spatial filtering operations we can perform is a smoothing operation

- Simply average all of the pixels in a neighbourhood around a central value
- Especially useful in removing noise from images Also useful for highlighting gross detail

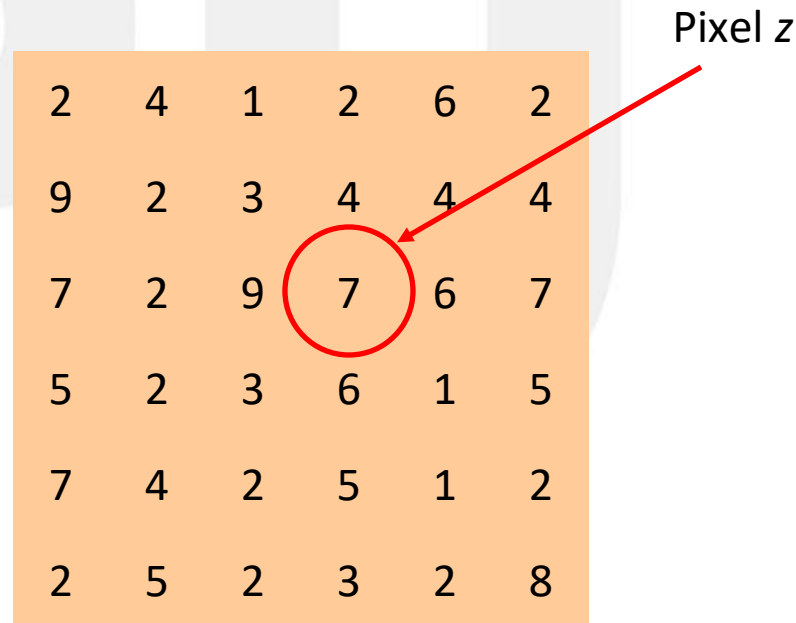
$1/9$	$1/9$	$1/9$
$1/9$	$1/9$	$1/9$
$1/9$	$1/9$	$1/9$

Simple averaging filter

6. Basics of Spatial Filtering

Sometime we need to manipulate values obtained from neighboring pixels

Example: How can we compute an average value of pixels in a 3x3 region center at a pixel z ?

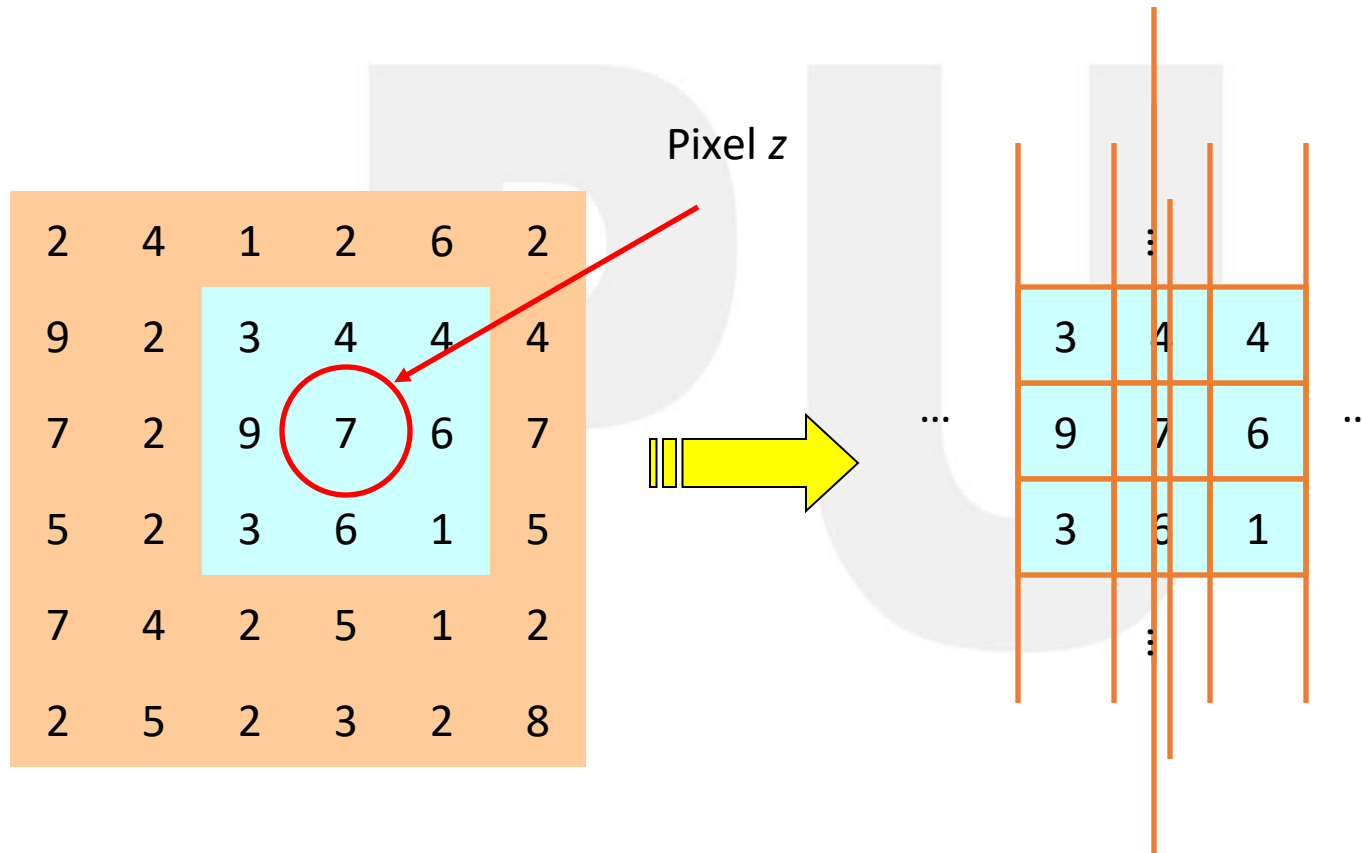


2	4	1	2	6	2
9	2	3	4	4	4
7	2	9	7	6	7
5	2	3	6	1	5
7	4	2	5	1	2
2	5	2	3	2	8

Image

Contd..

Step 1. Selected only needed pixels



Contd..

Step 2. Multiply every pixel by $\frac{1}{9}$ and then sum up the values

...

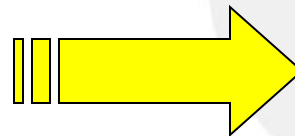
3	4	4
9	7	6
3	6	1

...

$\frac{1}{9}$

1	1	1
1	1	1
1	1	1

Mask or
Window or
Template



$$\begin{aligned}
 y &= \frac{1}{9} \cdot 3 + \frac{1}{9} \cdot 4 + \frac{1}{9} \cdot 4 \\
 &+ \frac{1}{9} \cdot 9 + \frac{1}{9} \cdot 7 + \frac{1}{9} \cdot 6 \\
 &+ \frac{1}{9} \cdot 3 + \frac{1}{9} \cdot 6 + \frac{1}{9} \cdot 1
 \end{aligned}$$

Contd..

How to compute the 3x3 average values at every pixels?

2	4	1	2	6	2
9	2	3	4	4	4
7	2	9	7	6	7
5	2	3	6	1	5
7	4	2	5	1	2

Solution: Imagine that we have a 3x3 window that can be placed everywhere on the image



Masking Window

Contd..

Step 1: Move the window to the first location where we want to compute the average value and then select only pixels inside the window.

2	4	1	2	6	2
9	2	3	4	4	4
7	2	9	7	6	7
5	2	3	6	1	5
7	4	2	5	1	2

Original image



2	4	1
9	2	3
7	2	9

Sub image p



Step 2: Compute the average value

$$y = \sum_{i=1}^3 \sum_{j=1}^3 \frac{1}{9} \cdot p(i, j)$$

Step 3: Place the result at the pixel in the output image

	4.		
	3		

Output image

Step 4: Move the window to the next location and go to Step 2

6. Basics of Spatial Filtering (cont.)

- The 3x3 averaging method is one example of the **mask operation** or **Spatial filtering**.
- The mask operation has the corresponding mask (sometimes called window or template).
- The mask contains coefficients to be multiplied with pixel values.

w(1, 1)	w(2, 1)	w(3, 1)
w(1, 2)	w(2, 2)	w(3, 2)
w(3, 1)	w(3, 2)	w(3, 3)

Mask coefficients

Example : moving averaging

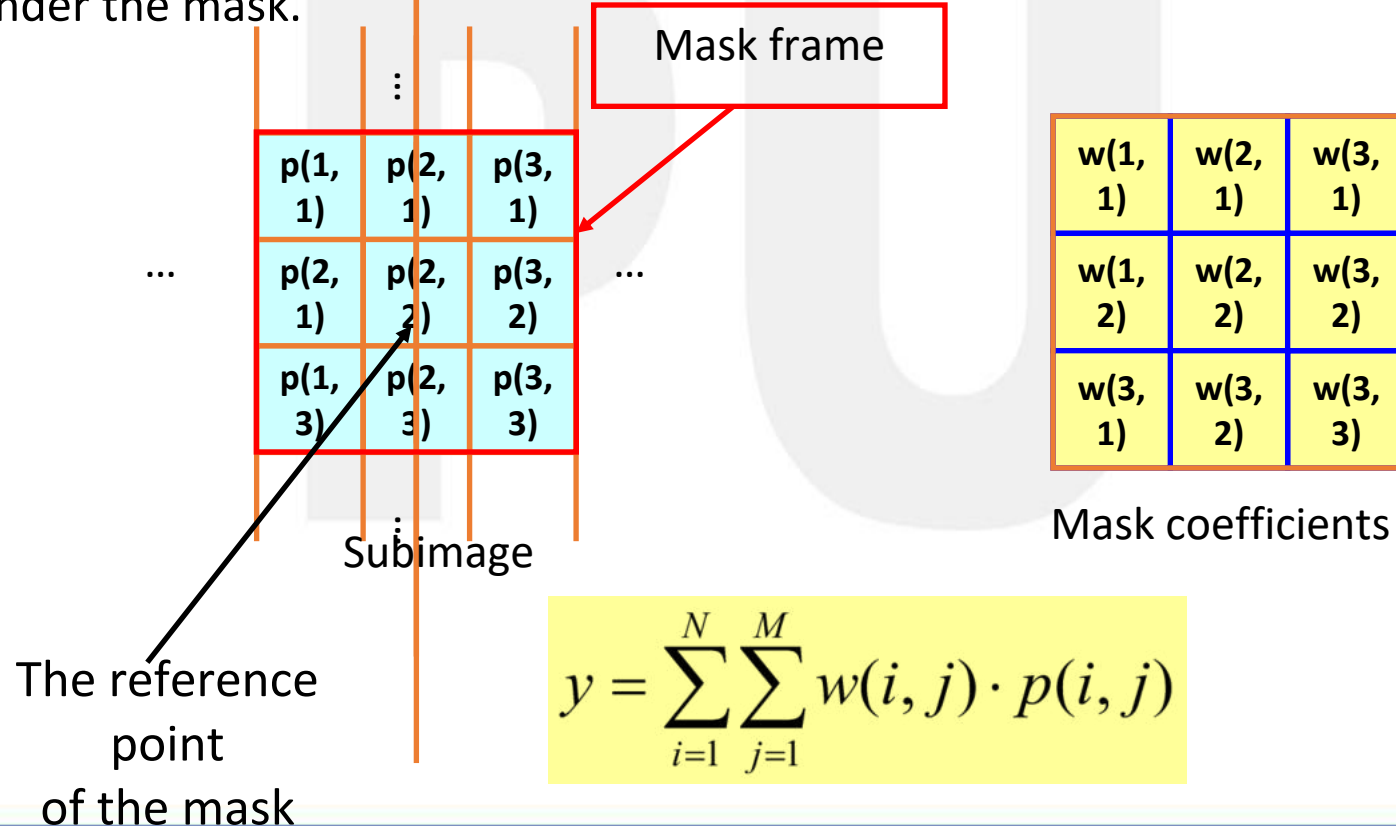
$\frac{1}{9}$	1	1	1
	1	1	1
	1	1	1

The mask of the 3x3 moving average filter has all coefficients = $1/9$

Contd..

The mask operation at each point is performed by:

1. Move the reference point (center) of mask to the location to be computed
2. Compute **sum of products** between mask coefficients and pixels in subimage under the mask.



Contd..

The spatial filtering on the whole image is given by:

1. Move the mask over the image at each location.
2. Compute sum of products between the mask coefficients and pixels inside subimage under the mask.
3. Store the results at the corresponding pixels of the output image.
4. Move the mask to the next location and go to step 2 until all pixel locations have been used.

Smoothing Spatial Filtering

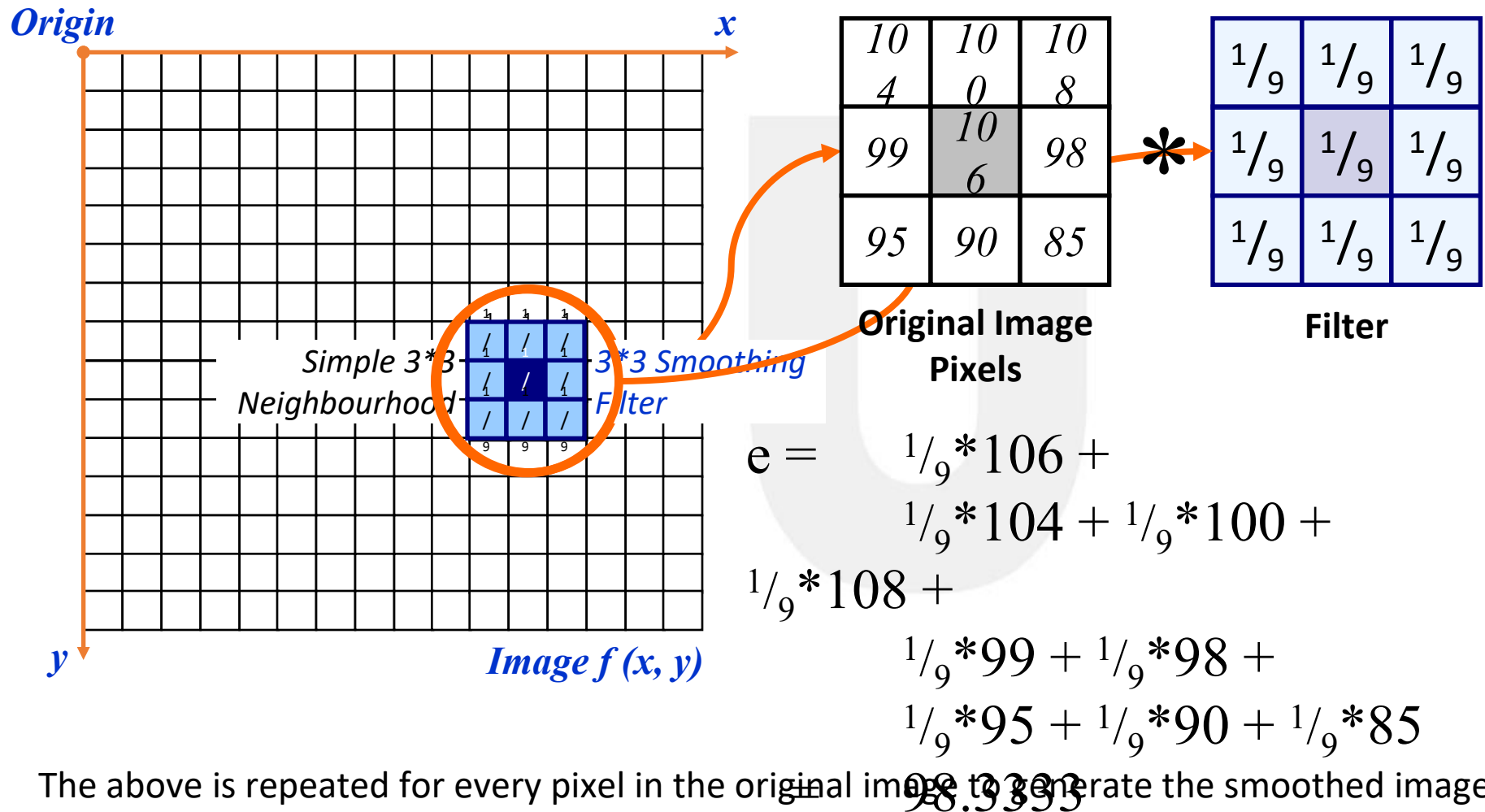


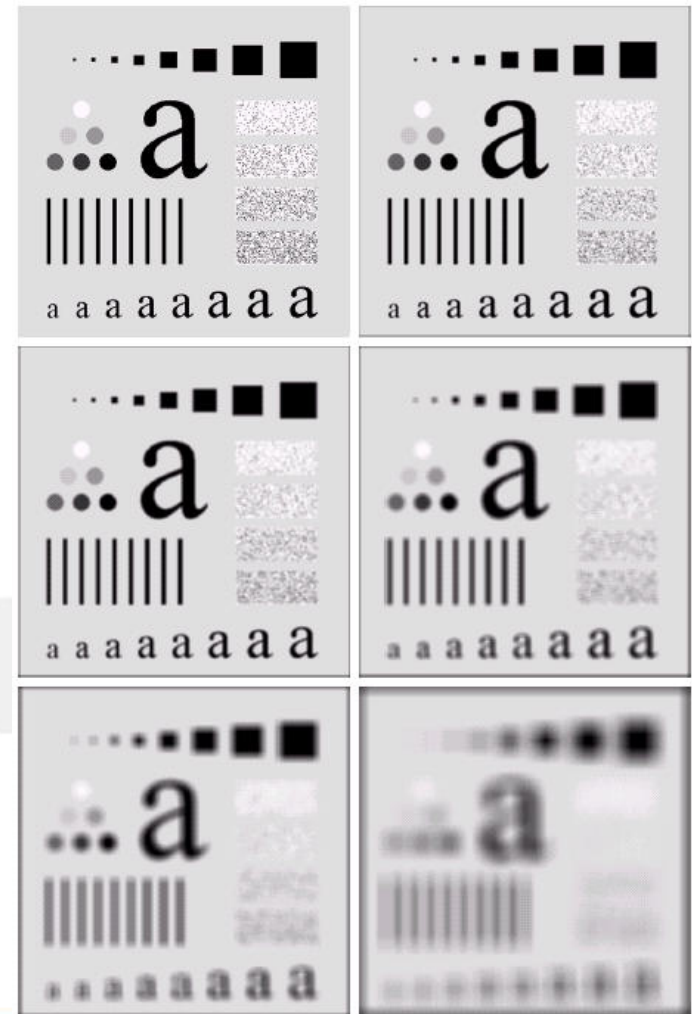
Image Smoothing Example

The image at the top left is an original image of size 500*500 pixels

The subsequent images show the image after filtering with an averaging filter of increasing sizes

- 3, 5, 9, 15 and 35

Notice how detail begins to disappear



Weighted Smoothing Filters

More effective smoothing filters can be generated by allowing different pixels in the neighbourhood different weights in the averaging function

- Pixels closer to the central pixel are more important
- Often referred to as a *weighted averaging*

$\frac{1}{16}$	$\frac{2}{16}$	$\frac{1}{16}$
$\frac{2}{16}$	$\frac{4}{16}$	$\frac{2}{16}$
$\frac{1}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

Weighted averaging filter

References

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